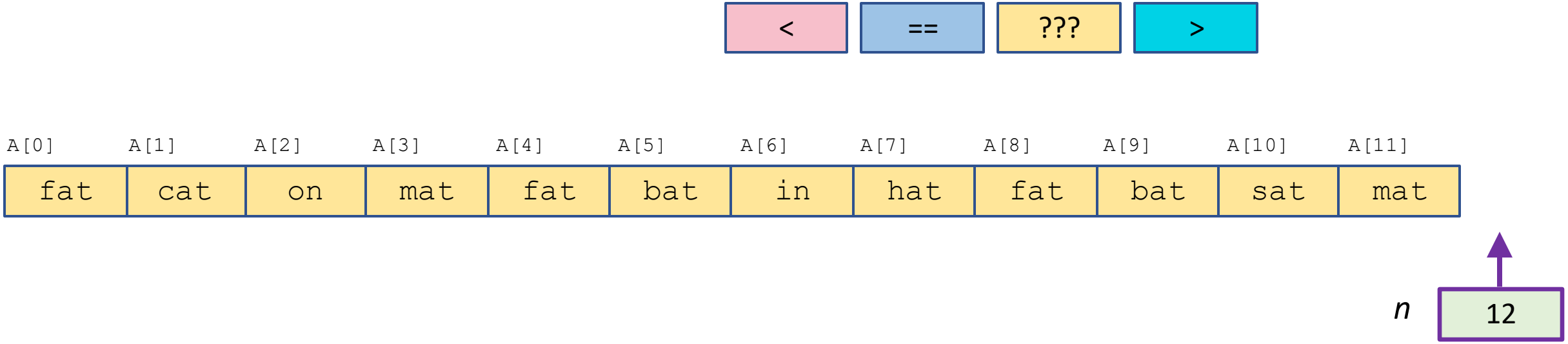
comp10002/comp20005
program quicksort.c

Support material for Chapter 12 of
Programming, Problem Solving, and Abstraction with C,
by Alistair Moffat

Prepared by Alistair Moffat, ammoffat@unimelb.edu.au


<
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???
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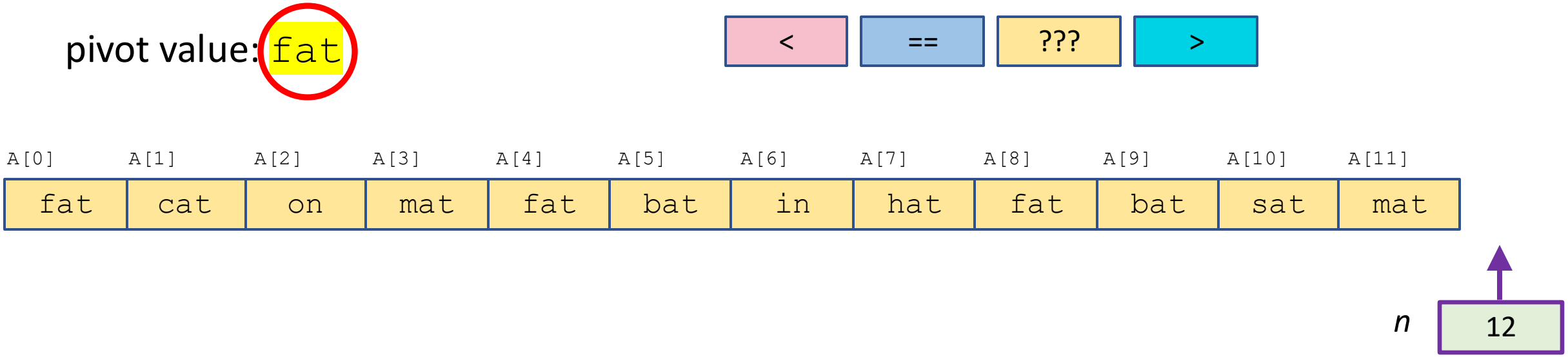
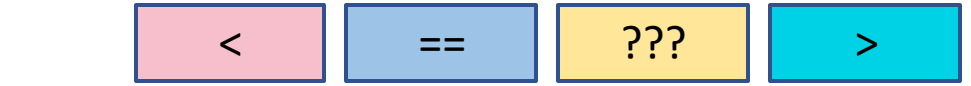
A[0]
A[1]
A[2]
A[3]
A[4]
A[5]
A[6]
A[7]
A[8]
A[9]
A[10]
A[11]
fat
cat
on
mat
fat
bat
in
hat
fat
bat
sat
mat
n
12

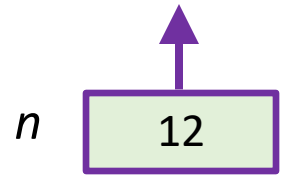
pivot value: `fat`



pivot value: `fat`



A[0]	A[1]	A[2]	A[3]	A[4]	A[5]	A[6]	A[7]	A[8]	A[9]	A[10]	A[11]
fat	cat	on	mat	fat	bat	in	hat	fat	bat	sat	mat



perform a three-way partition...

pivot value: fat

A[0]	A[1]	A[2]	A[3]	A[4]	A[5]	A[6]	A[7]	A[8]	A[9]	A[10]	A[11]
fat	cat	on	mat	fat	bat	in	hat	fat	bat	sat	mat

n 12

perform a three-way partition...

A[0]	A[1]	A[2]	A[3]	A[4]	A[5]	A[6]	A[7]	A[8]	A[9]	A[10]	A[11]
cat	bat	bat	fat	fat	fat	hat	in	mat	sat	mat	on

3 *fe*

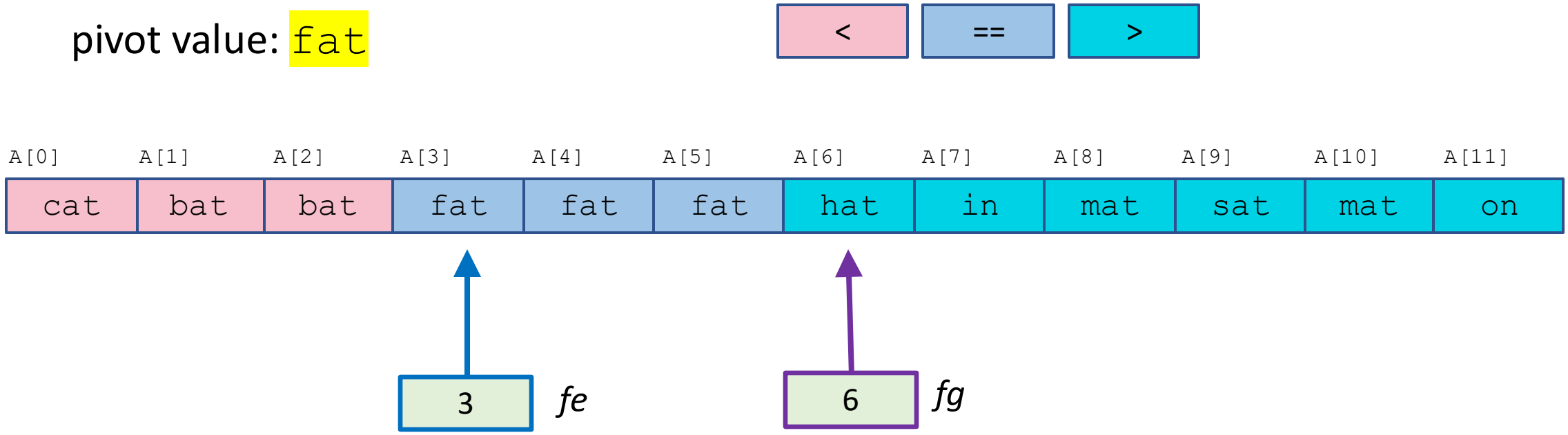
6 *fg*

hmmm, ok, maybe;
then what happens?

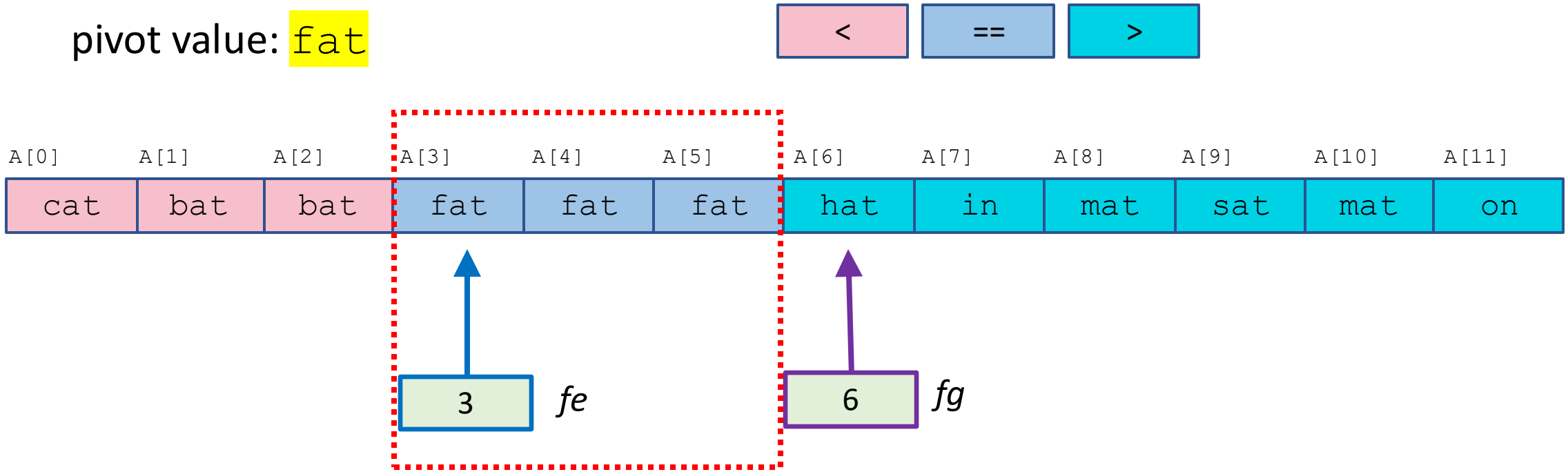
hmmm, ok, maybe;
then what happens?

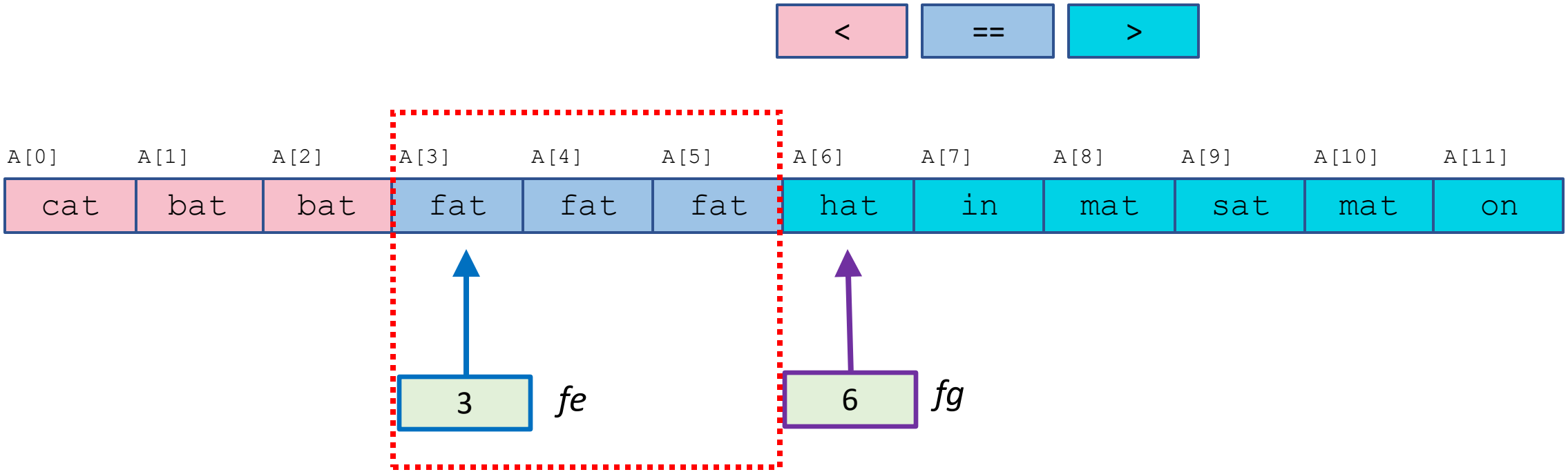
invisible friends!

pivot value: `fat`

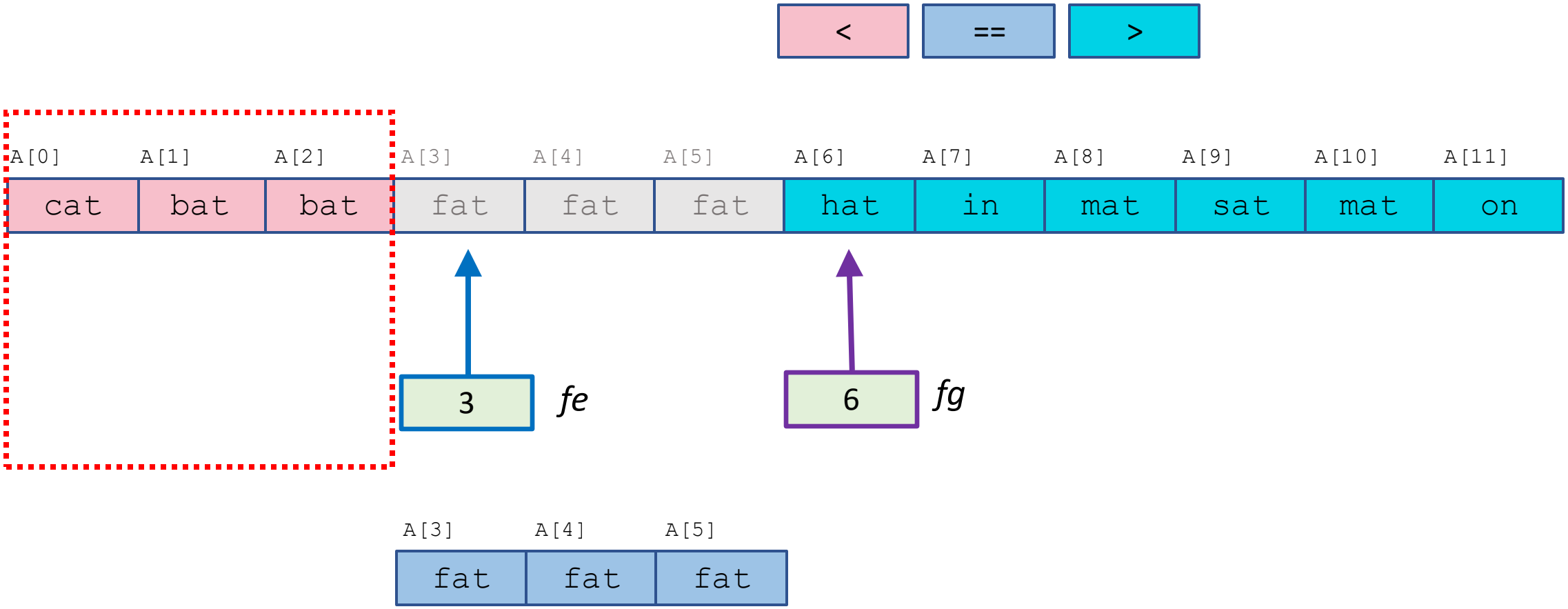


pivot value: `fat`





must be in correct locations, don't need to move again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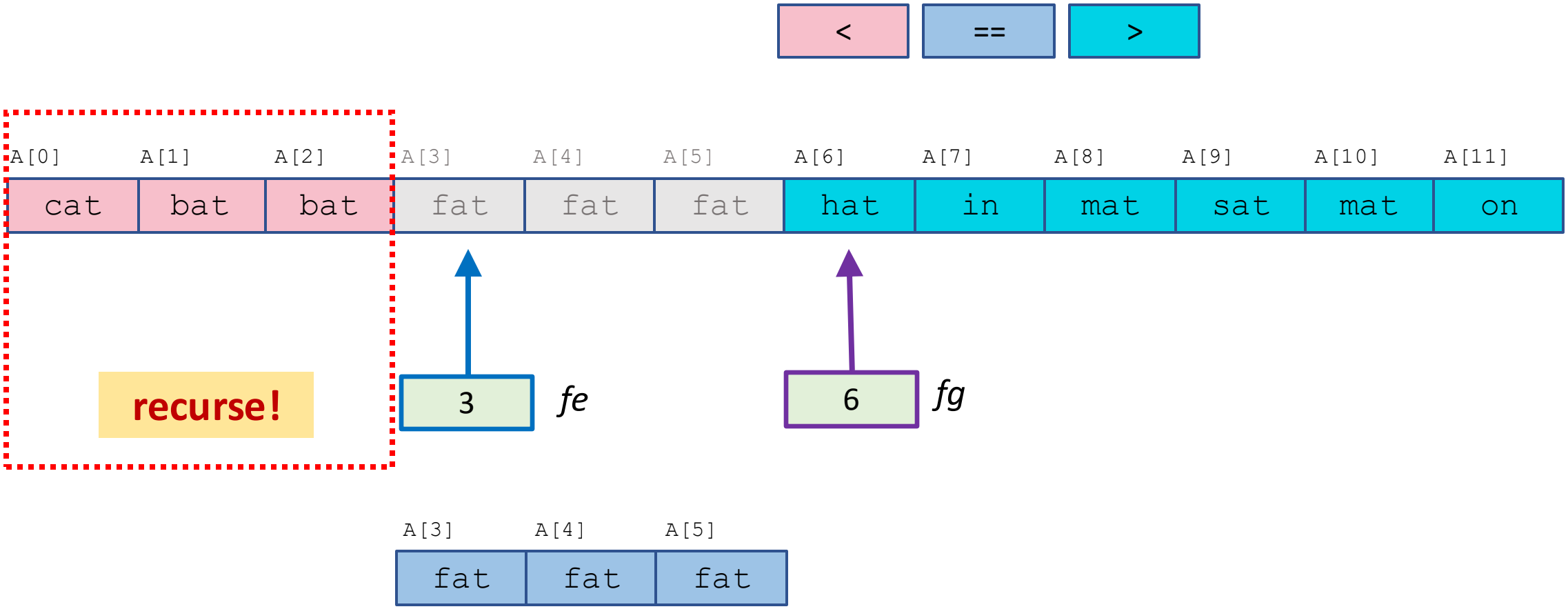
A[0]
A[1]
A[2]
A[3]
A[4]
A[5]
A[6]
A[7]
A[8]
A[9]
A[10]
A[11]
cat
bat
bat
fat
fat
fat
hat
in
mat
sat
mat
on
3
fe
6
fg

A[3]
A[4]
A[5]
fat
fat
fat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A[0]
A[1]
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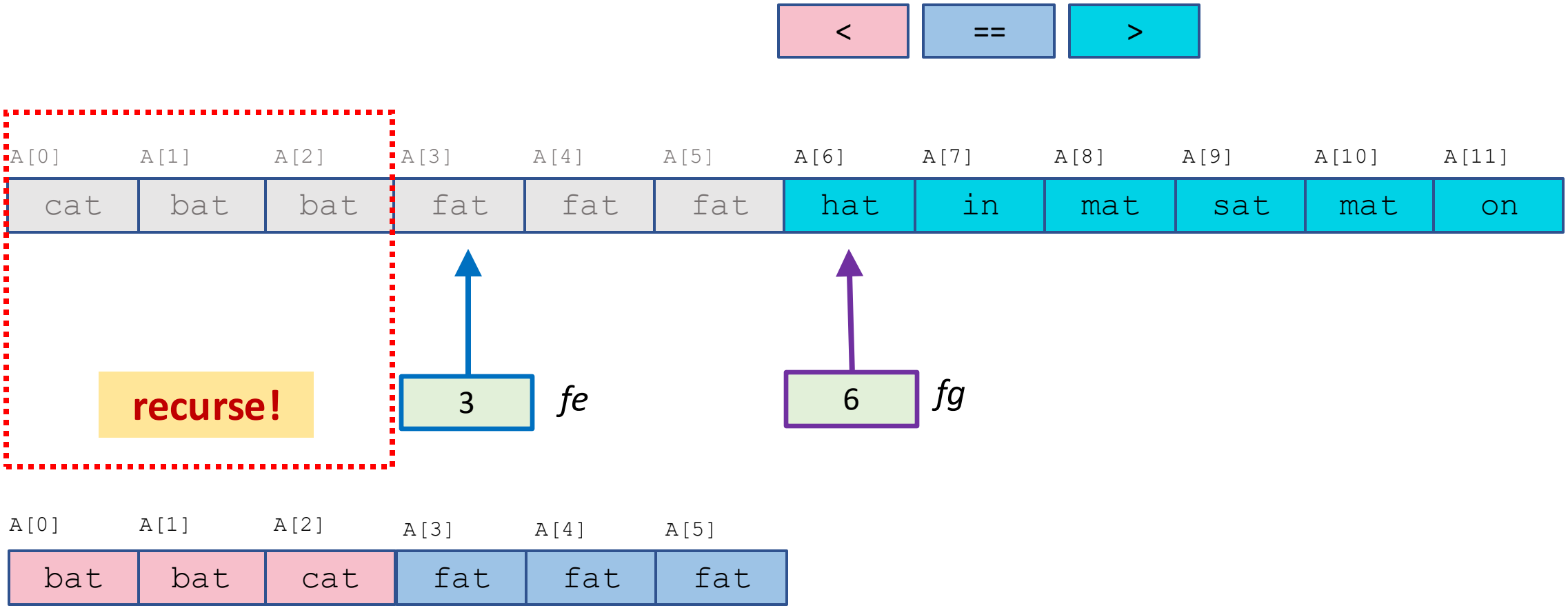
A[6]
A[7]
A[8]
A[9]
A[10]
A[11]
cat
bat
bat
fat
fat
fat
hat
in
mat
sat
mat
on
recurse!
3
fe
6
fg

A[3]
A[4]
A[5]
fat
fat
fat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A[0]
A[1]
A[2]
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A[11]
cat
bat
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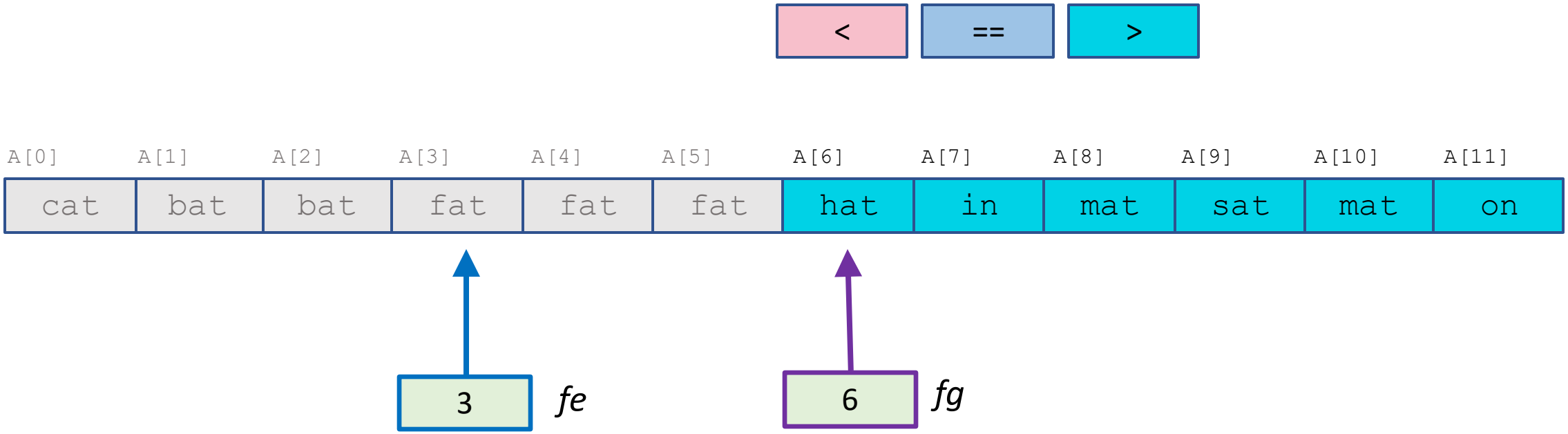
fat
fat
fat
hat
in
mat
sat
mat
on
recurse!
3
fe
6
fg
A[0]
A[1]
A[2]
A[3]
A[4]
A[5]
bat
bat
cat
fat
fat
fat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A[0]
A[1]
A[2]
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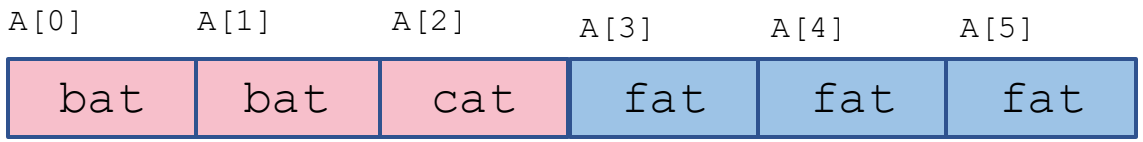
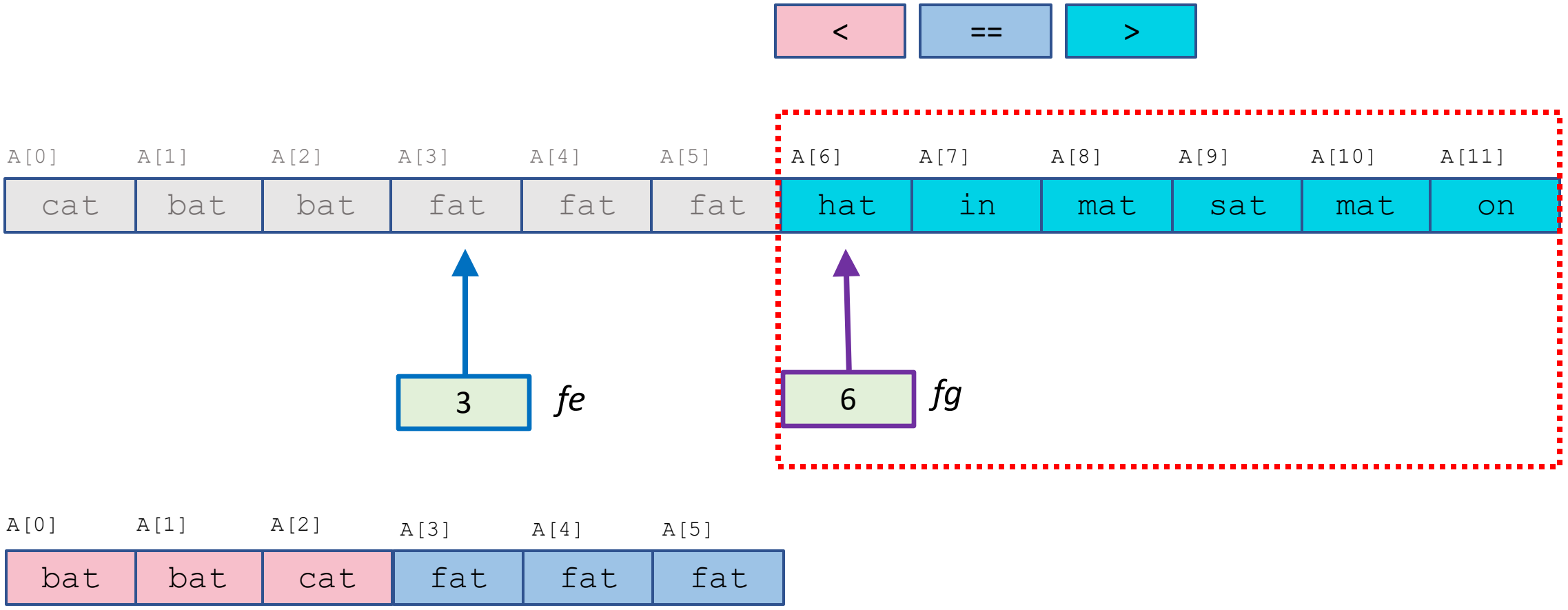
A[5]
A[6]
A[7]
A[8]
A[9]
A[10]
A[11]
cat
bat
bat
fat
fat
fat
hat
in
mat
sat
mat
on
3
fe
6
fg


A[0]
A[1]
A[2]
A[3]
A[4]
A[5]
bat
bat
cat
fat
fat
fat


A[0]
A[1]
A[2]
A[3]
A[4]
A[5]
A[6]
A[7]
A[8]
A[9]
A[10]
A[11]
cat
bat
bat
fat
fat
fat
hat
in
mat
sat
mat
on
3
fe
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fg
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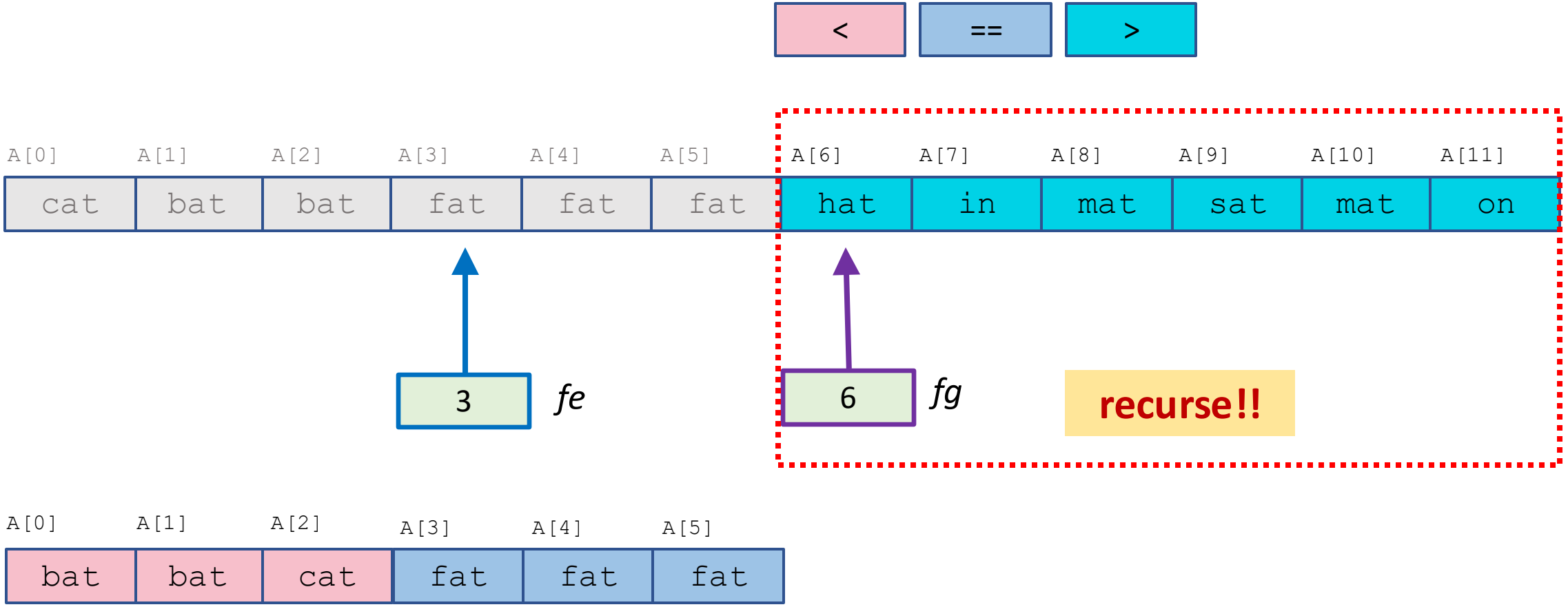
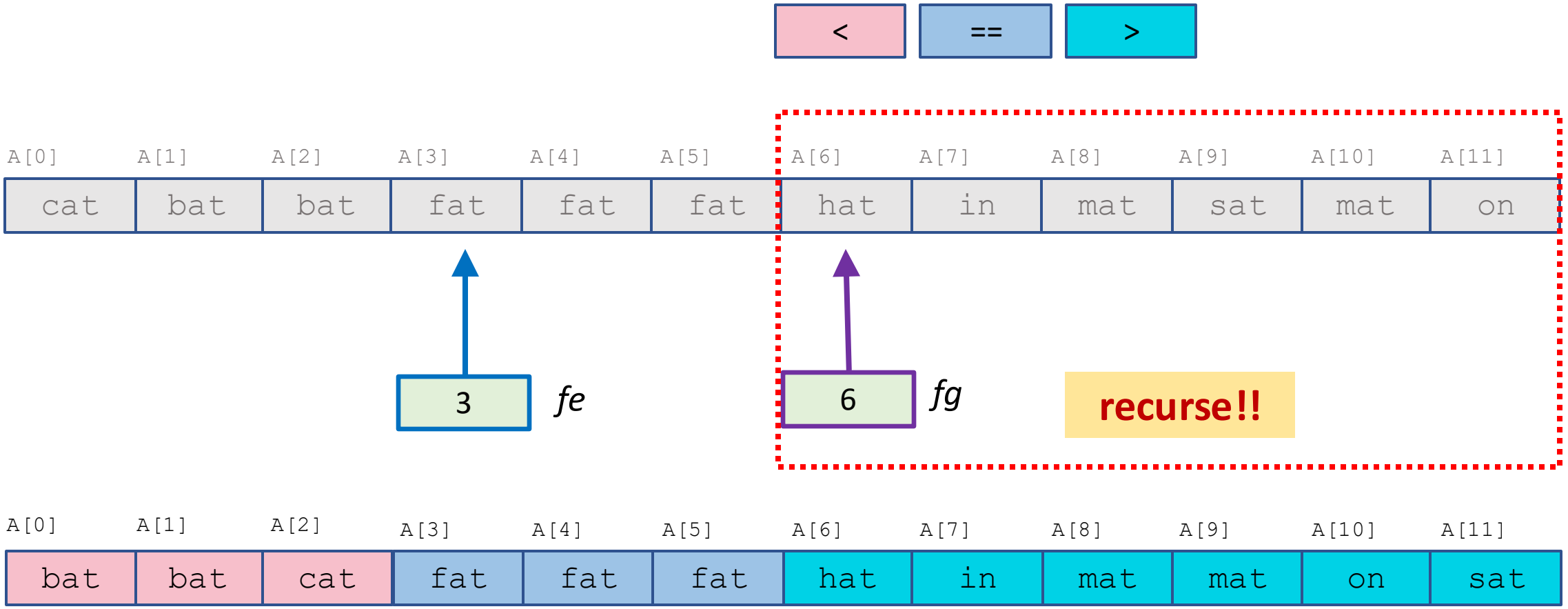
==
>
A[0]
A[1]
A[2]
A[3]
A[4]
A[5]
bat
bat
cat
fat
fat
fat


A[0]
A[1]
A[2]
A[3]
A[4]
A[5]
A[6]
A[7]
A[8]
A[9]
A[10]
A[11]
cat
bat
bat
fat
fat
fat
hat
in
mat
sat
mat
on
3
fe
6
fg
recurse!!

A[0]
A[1]
A[2]
A[3]
A[4]
A[5]
bat
bat
cat
fat
fat
fat


<
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A[0]
A[1]
A[2]
A[3]
A[4]
A[5]
A[6]
A[7]
A[8]
A[9]
A[10]
A[11]
cat
bat
bat
fat
fat
fat
hat
in
mat
sat
mat
on
3
fe
6
fg
recurse!!
A[0]
A[1]
A[2]
A[3]
A[4]
A[5]
A[6]
A[7]
A[8]
A[9]
A[10]
A[11]
bat
bat
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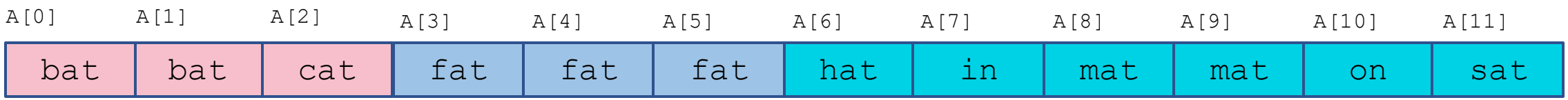
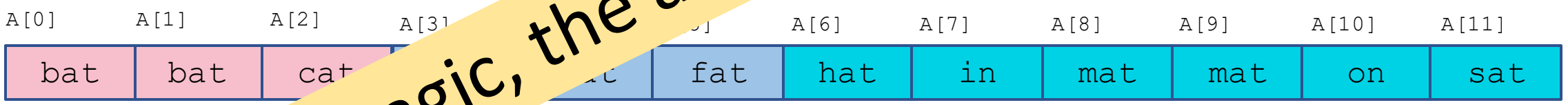
fat
fat
hat
in
mat
mat
on
sat


A[0]
A[1]
A[2]
A[3]
A[4]
A[5]
A[6]
A[7]
A[8]
A[9]
A[10]
A[11]
bat
bat
cat
fat
fat
fat
hat
in
mat
mat
on
sat



magic, the array is sorted

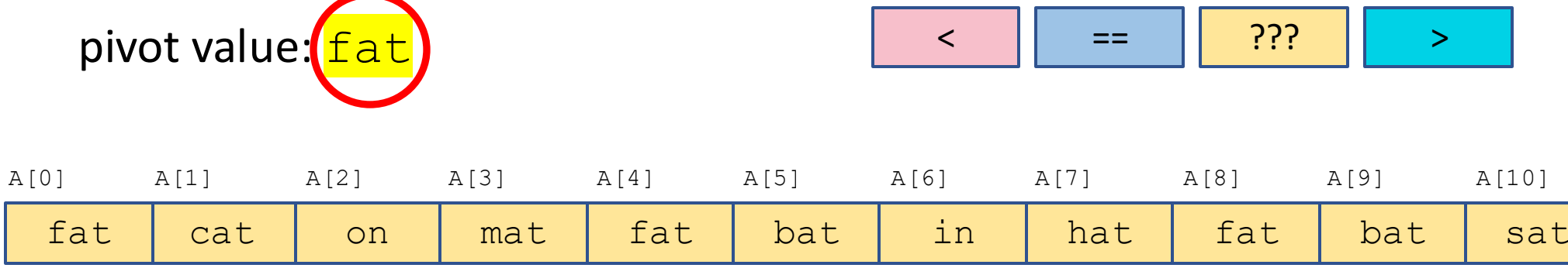
magic, the array is sorted

thanks, invisible friends!

hang on a minute!!

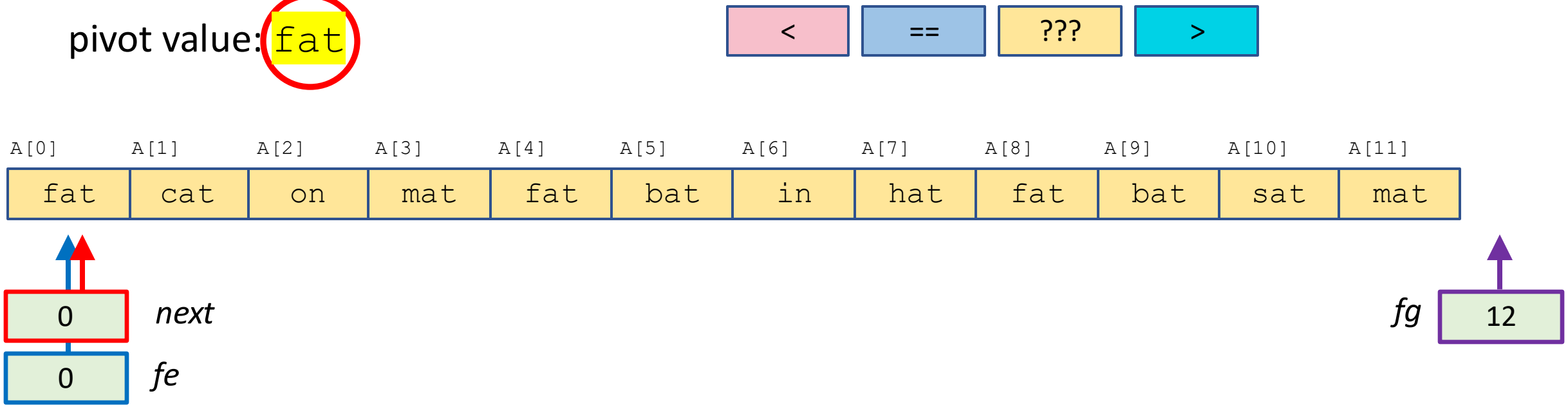
how does the three-way partition get done, how easy is that?

pivot value: `fat`



next, *fe*, *fg* ← 0, 0, *n*
while *next* < *fg* **do**
  if *A*[*next*] < *p* **then**
    swap *A*[*fe*] and *A*[*next*]
    fe, *next* ← *fe* + 1, *next* + 1
  else if *A*[*next*] > *p* **then**
    swap *A*[*next*] and *A*[*fg* − 1]
    fg ← *fg* − 1
  else
    next ← *next* + 1

pivot value: `fat`



< == ??? >

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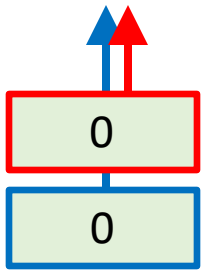
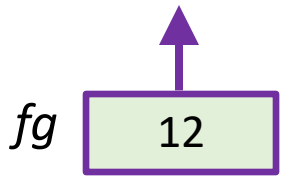
  else if *A*[*next*] > *p* **then**
    swap *A*[*next*] and *A*[*fg* − 1]
    fg ← *fg* − 1
  else
    next ← *next* + 1

pivot value: `fat`

<	==	???	>

A[0]	A[1]	A[2]	A[3]	A[4]	A[5]	A[6]	A[7]	A[8]	A[9]	A[10]	A[11]
fat	cat	on	mat	fat	bat	in	hat	fat	bat	sat	mat





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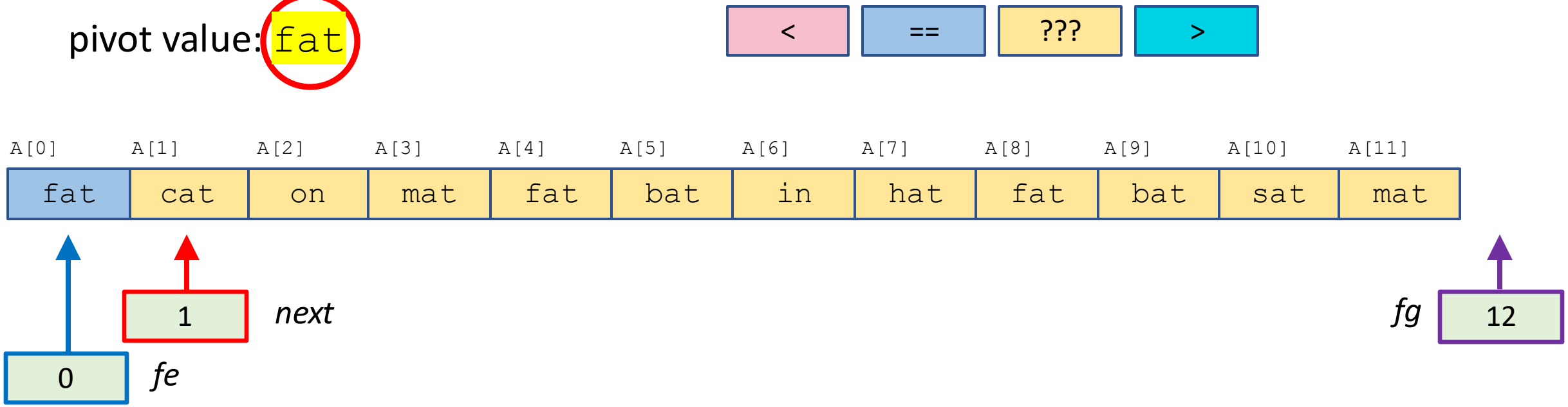


```
next, fe, fg ← 0, 0, n
while next < fg do
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        fe, next ← fe + 1, next + 1
    else if A[next] > p then
        swap A[next] and A[fg − 1]
        fg ← fg − 1
    else
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```
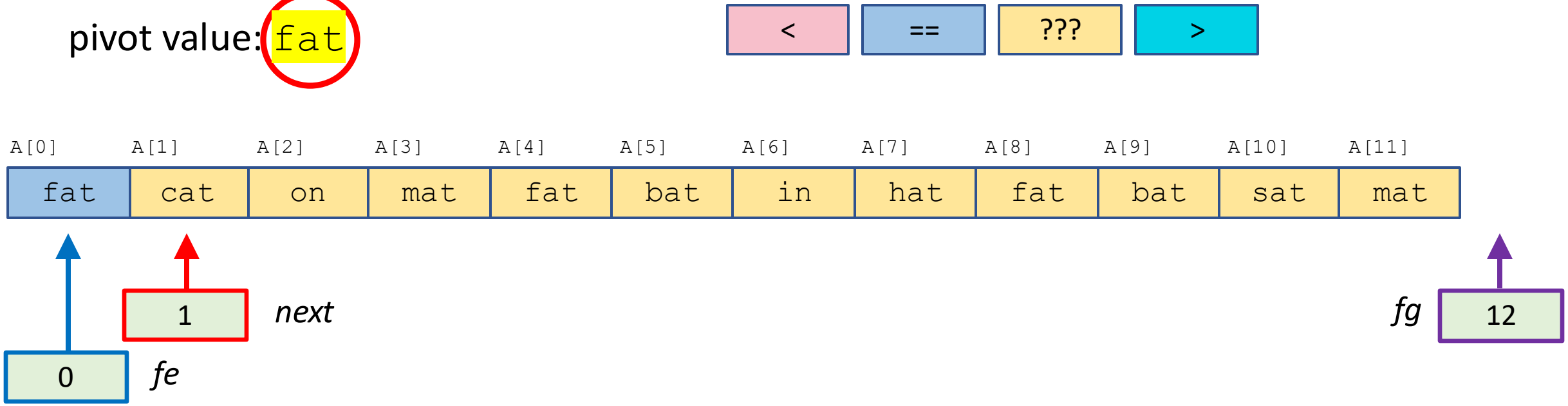

pivot value: `fat`



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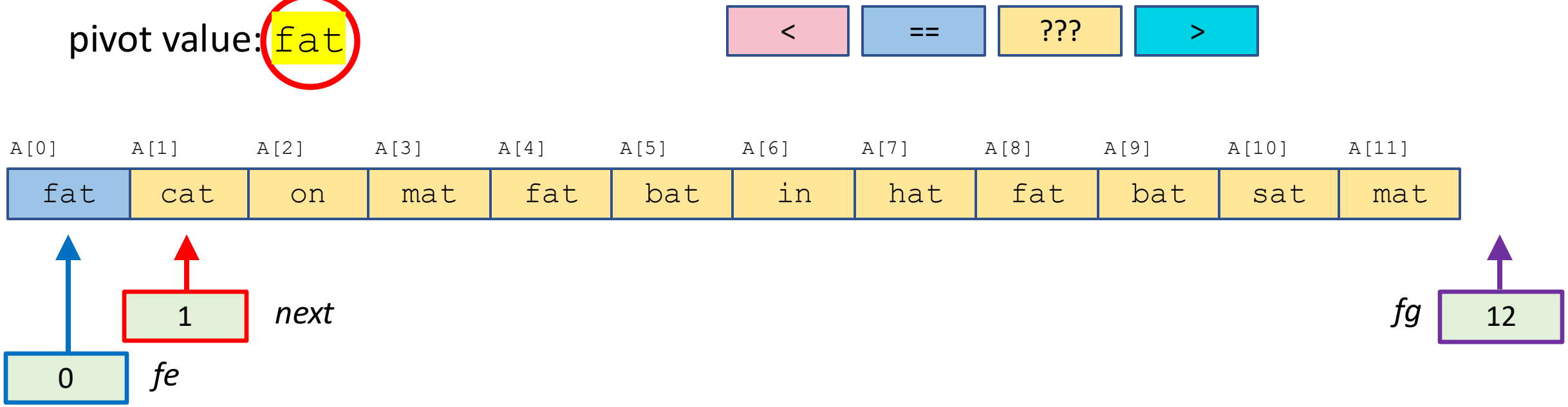
pivot value: `fat`



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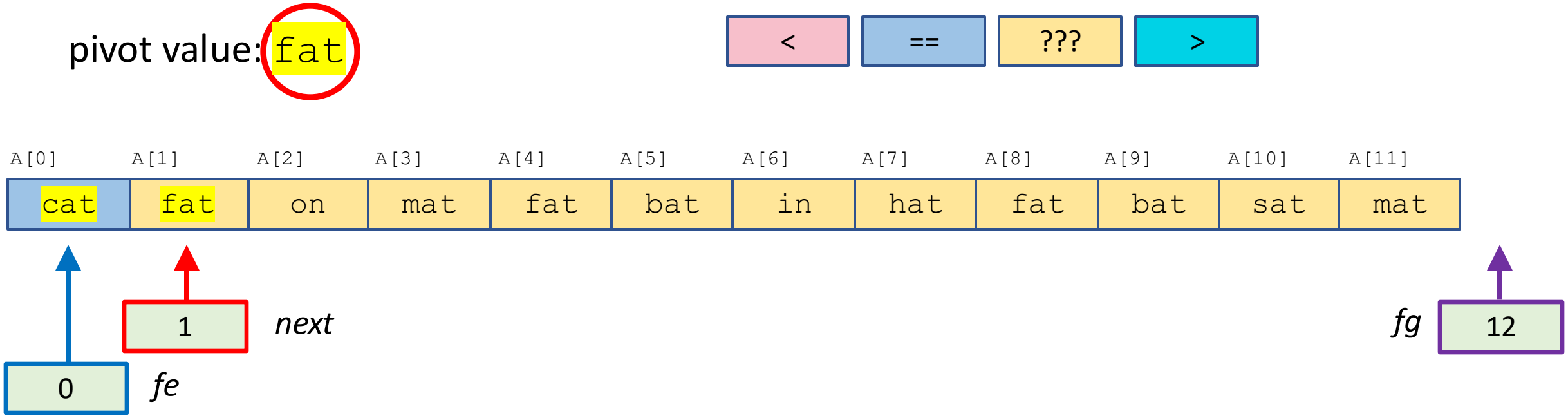
pivot value: fat



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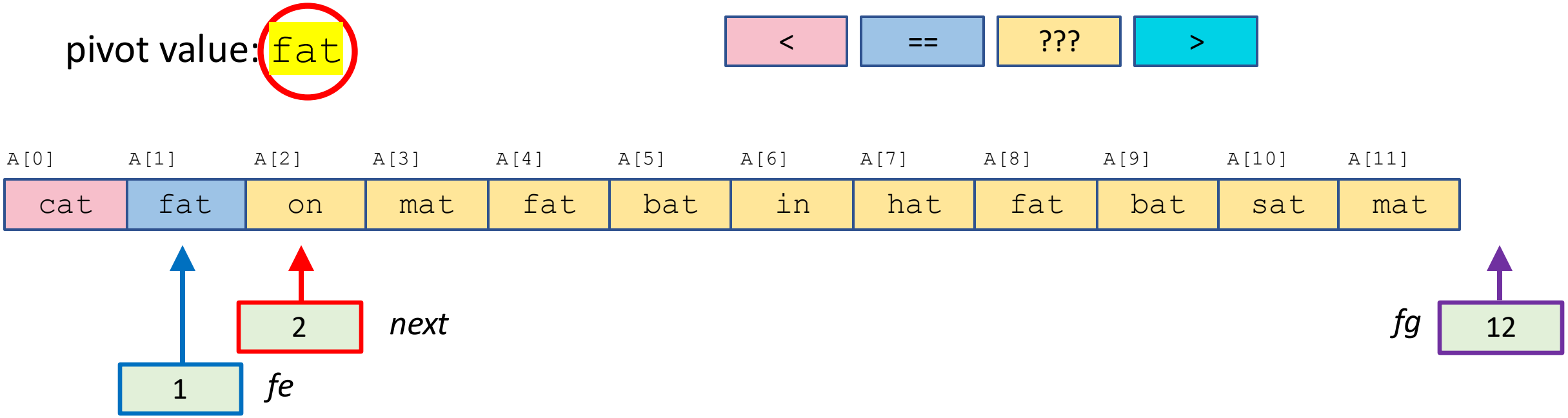
pivot value: fat

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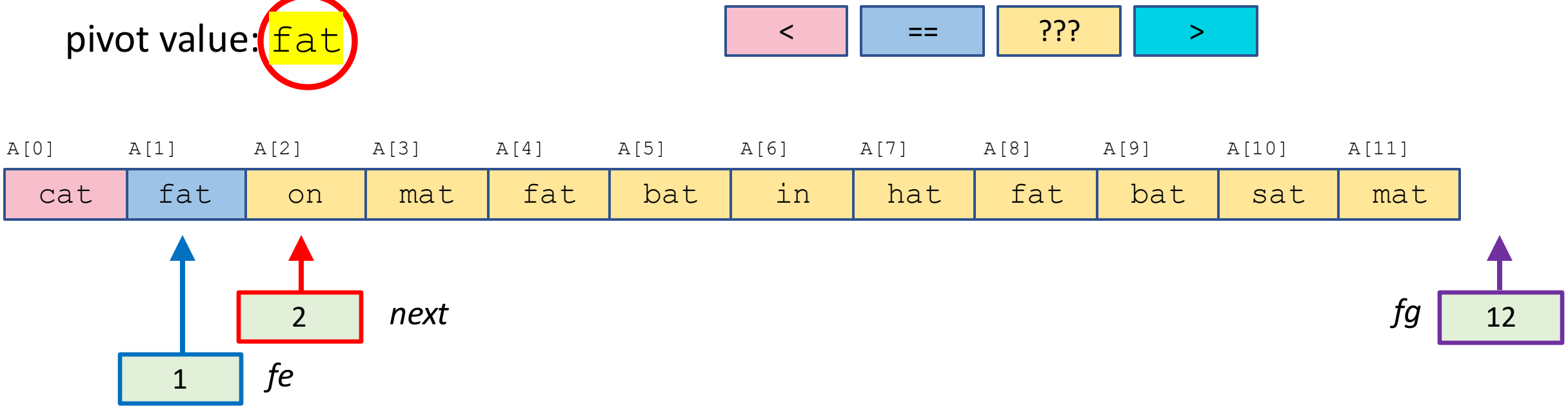
pivot value: fat



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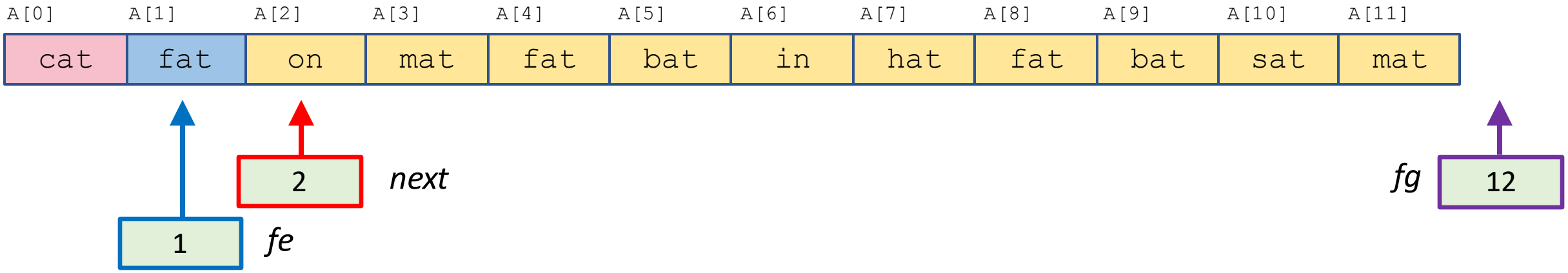
    next ← *next* + 1

pivot value: fat



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    fg ← *fg* − 1
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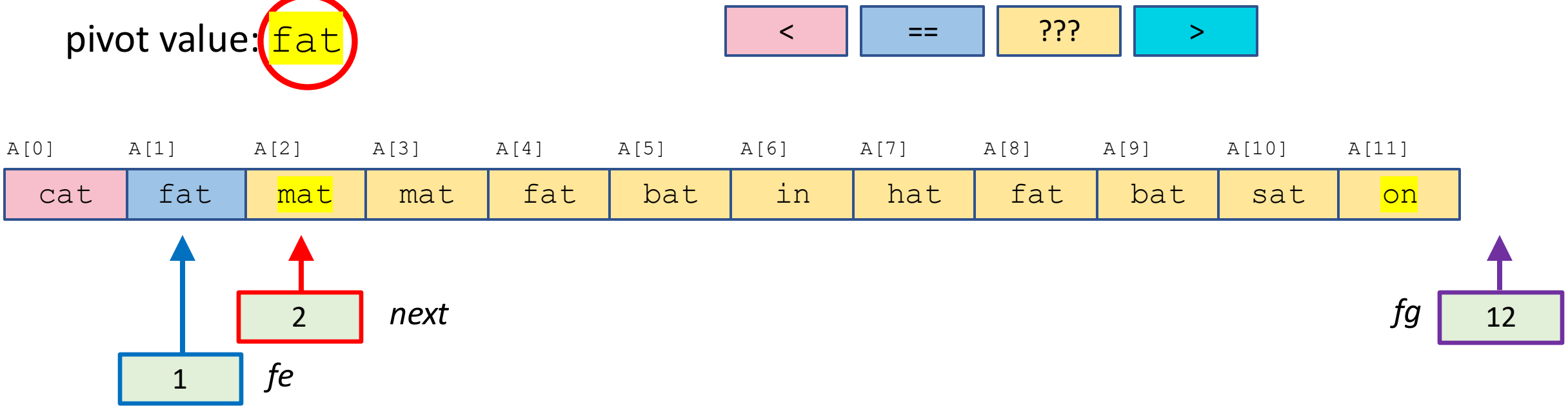

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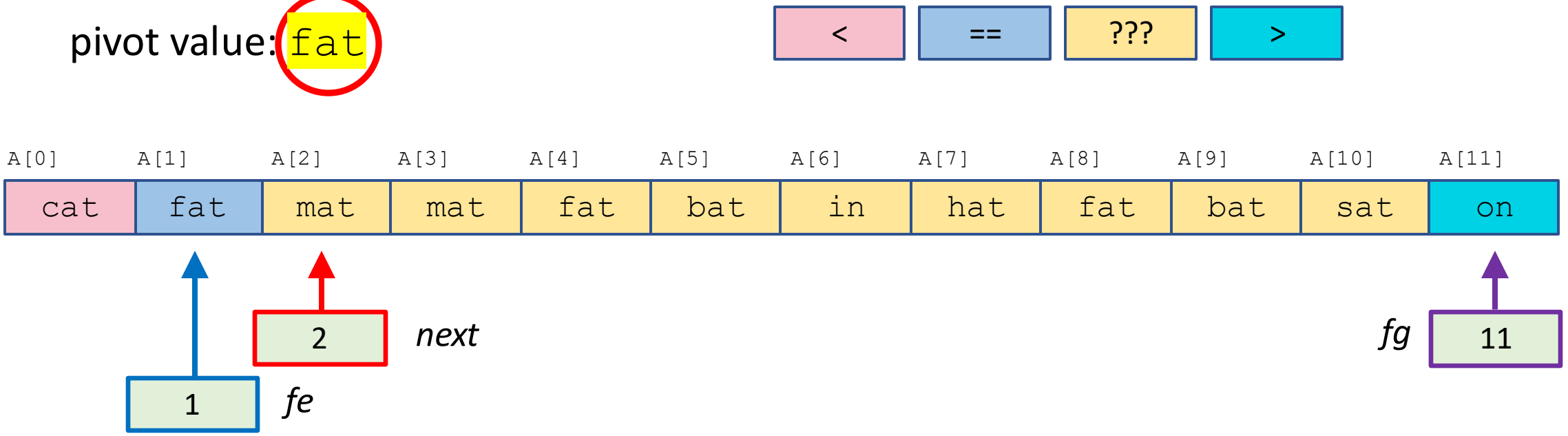
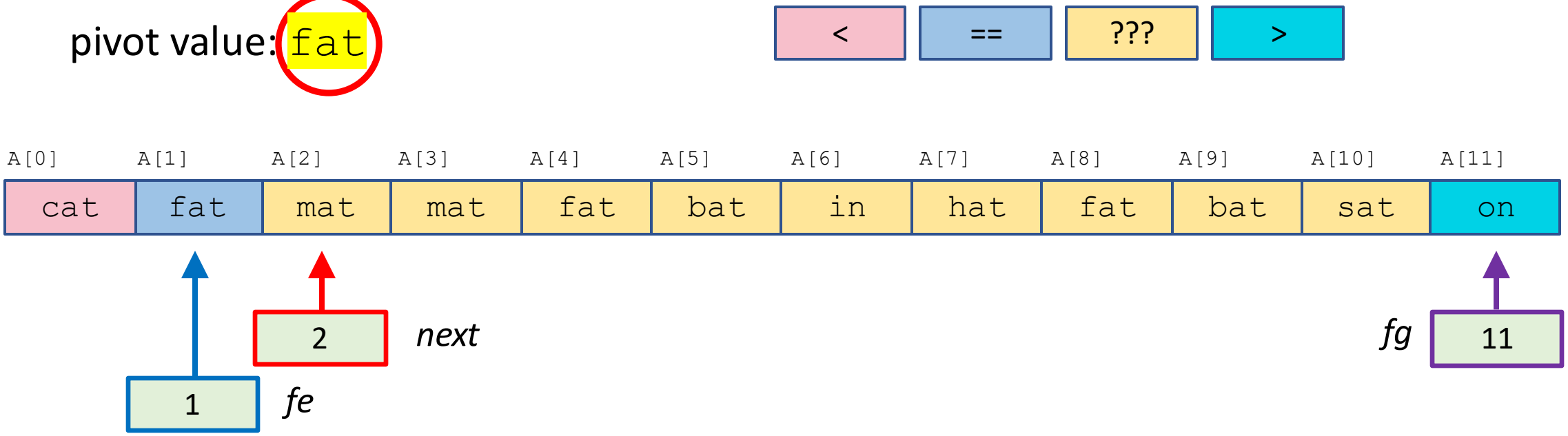
    fg ← *fg* − 1
  else
    next ← *next* + 1

pivot value: fat



next, *fe*, *fg* ← 0, 0, *n*
while *next* < *fg* **do**
  if *A*[*next*] < *p* **then**
    swap *A*[*fe*] and *A*[*next*]
    fe, *next* ← *fe* + 1, *next* + 1
  else if *A*[*next*] > *p* **then**
    swap *A*[*next*] and *A*[*fg* − 1]
    fg ← *fg* − 1
  else
    next ← *next* + 1

pivot value: fat



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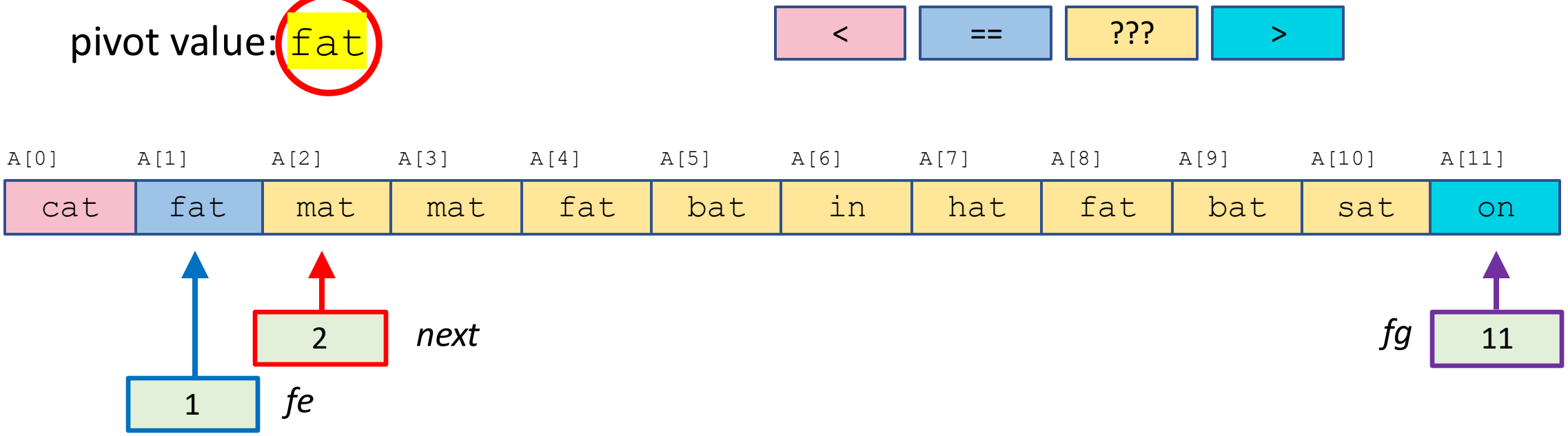

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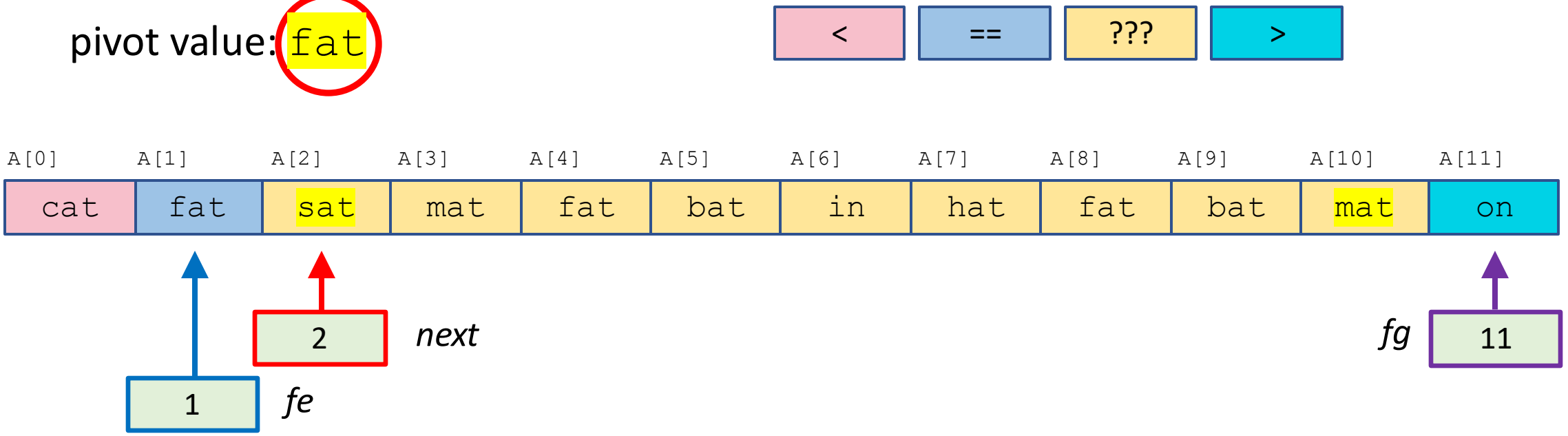

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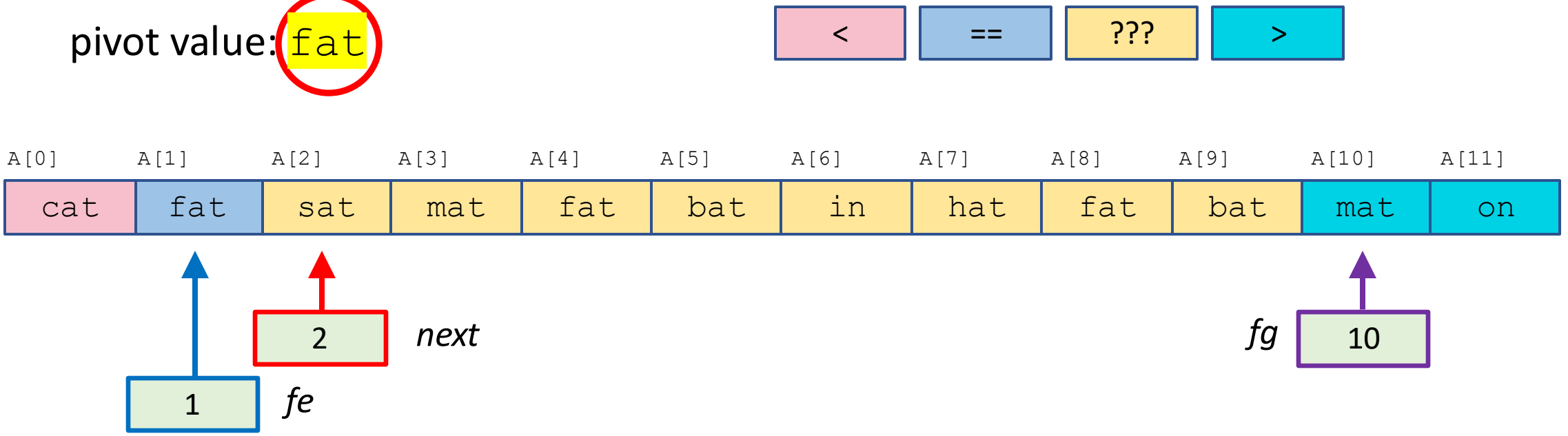

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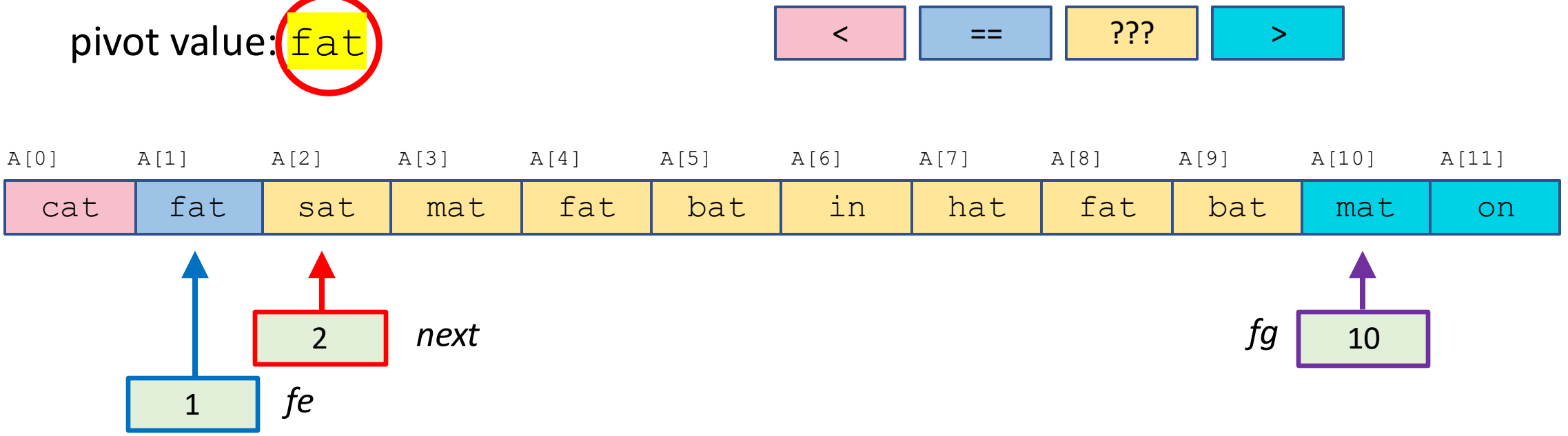

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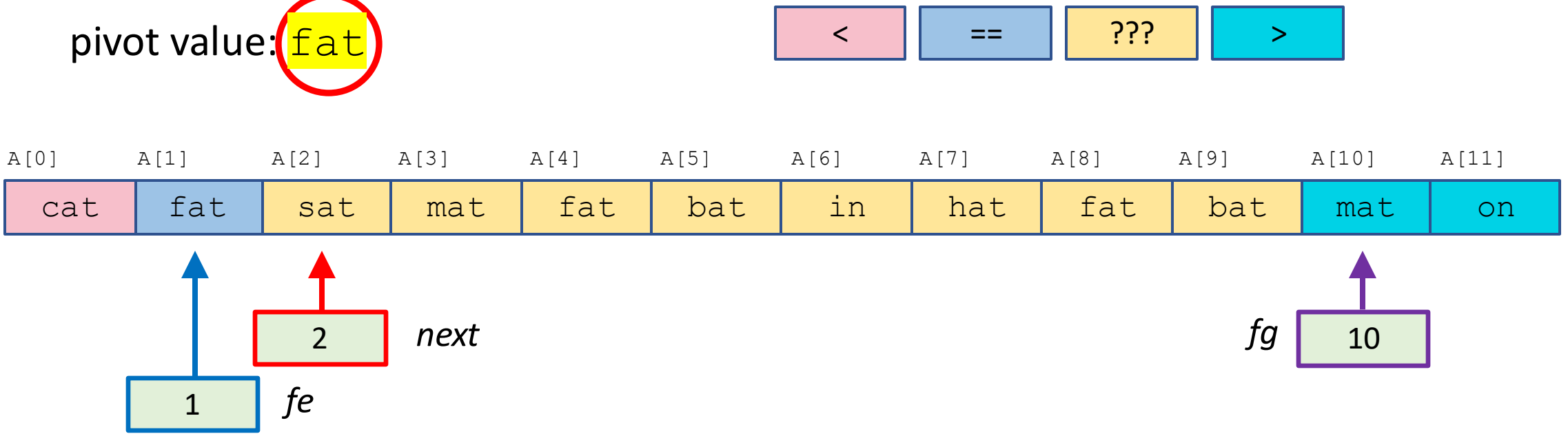

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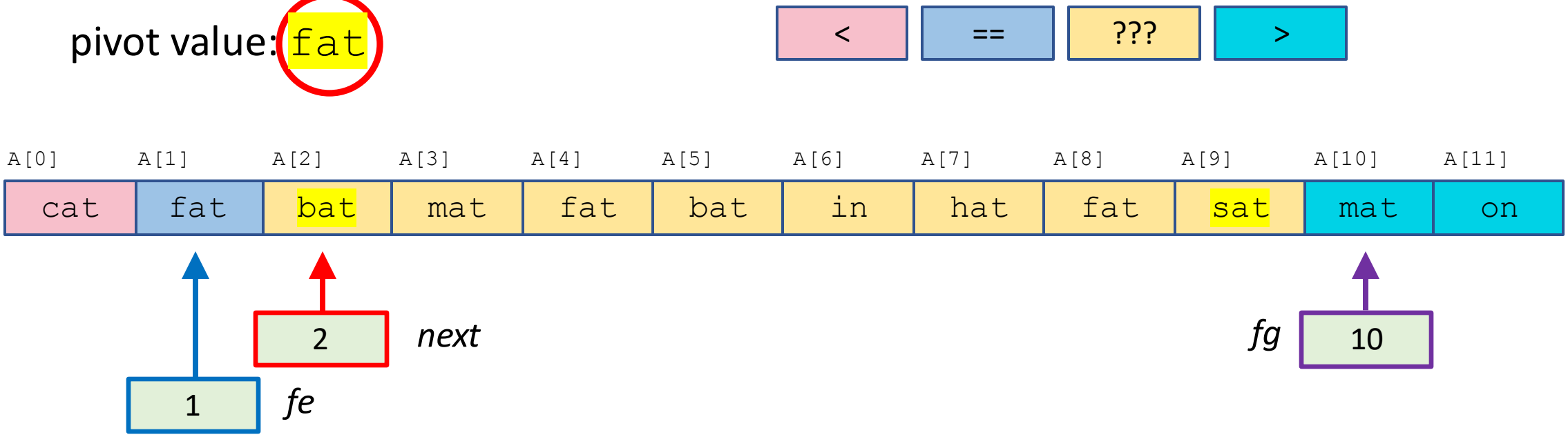

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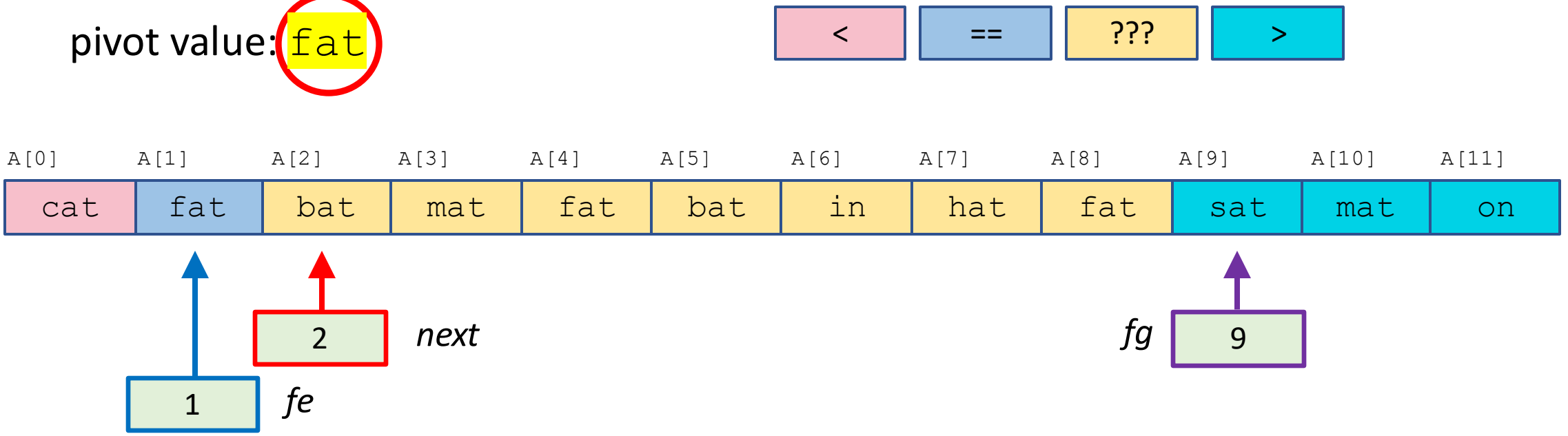


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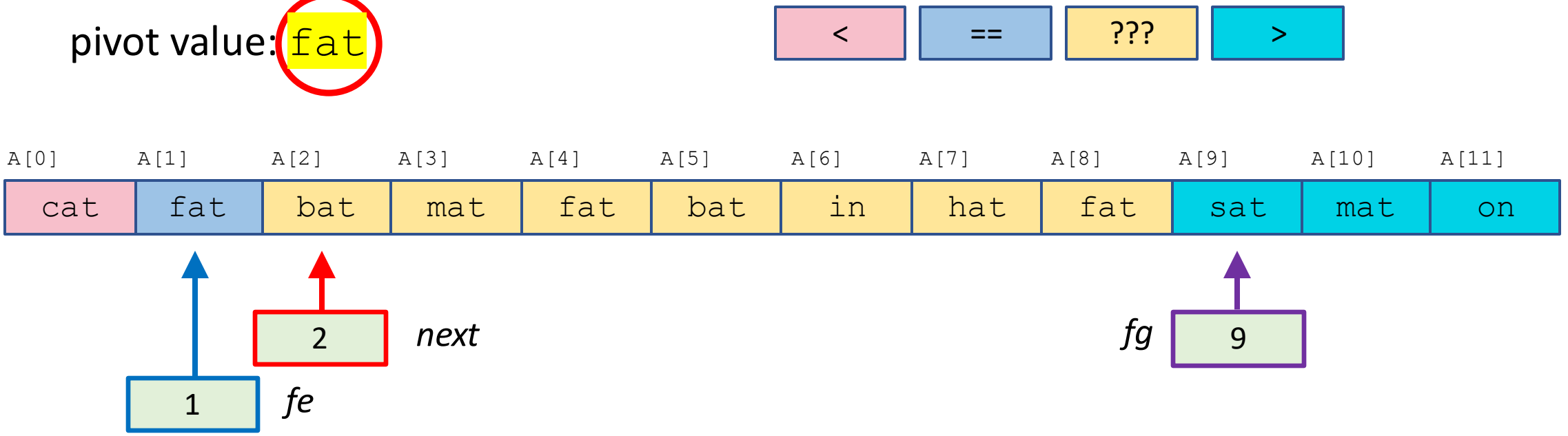
pivot value: fat



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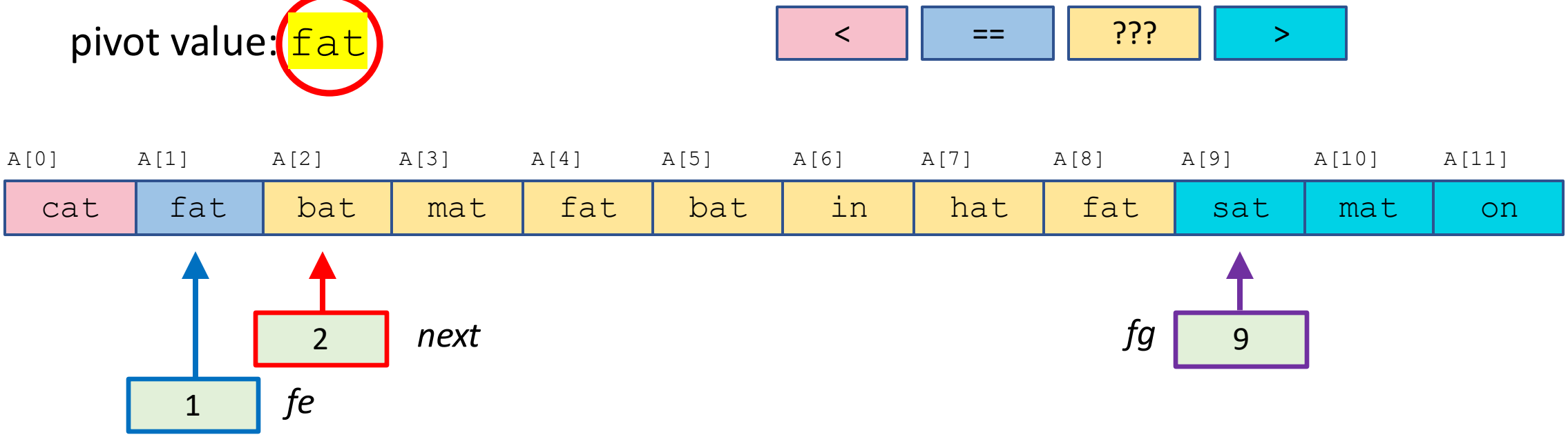


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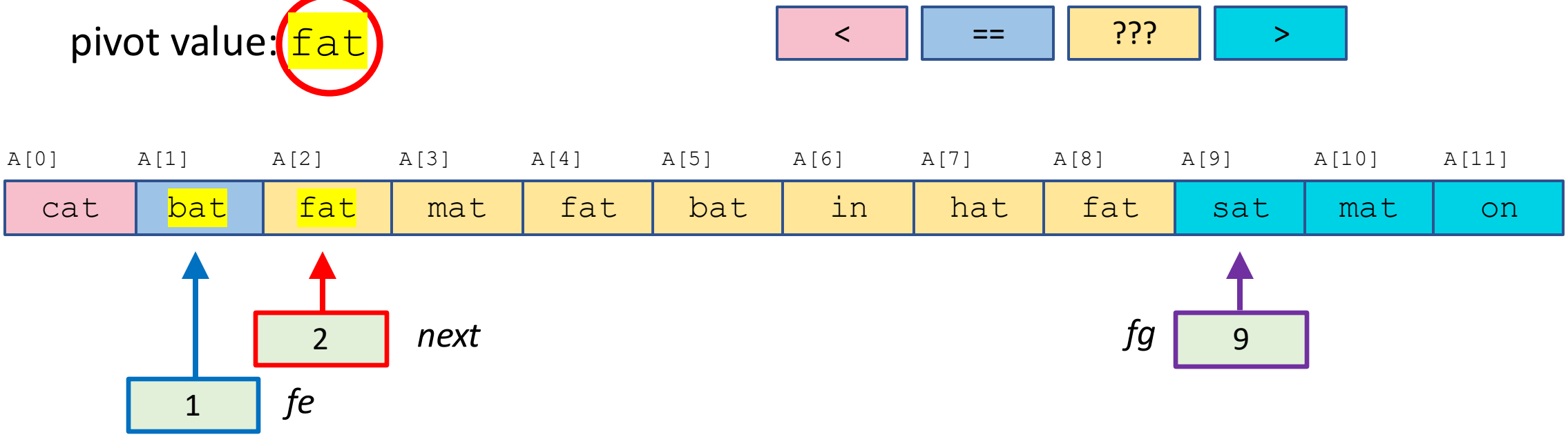

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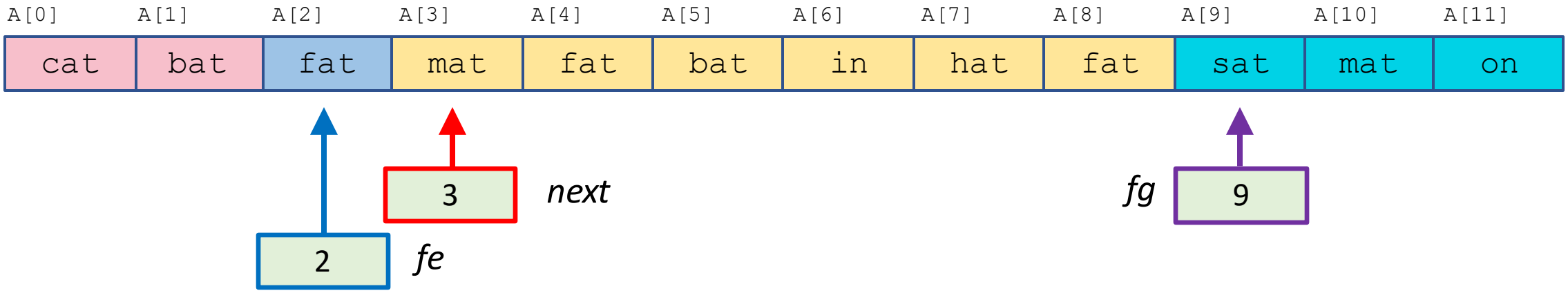


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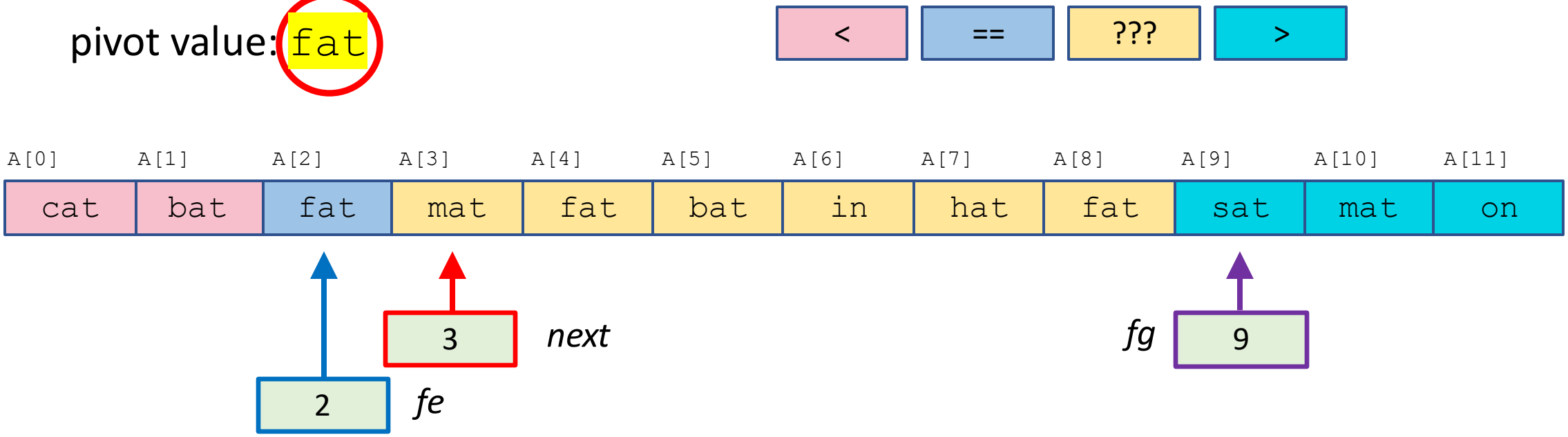

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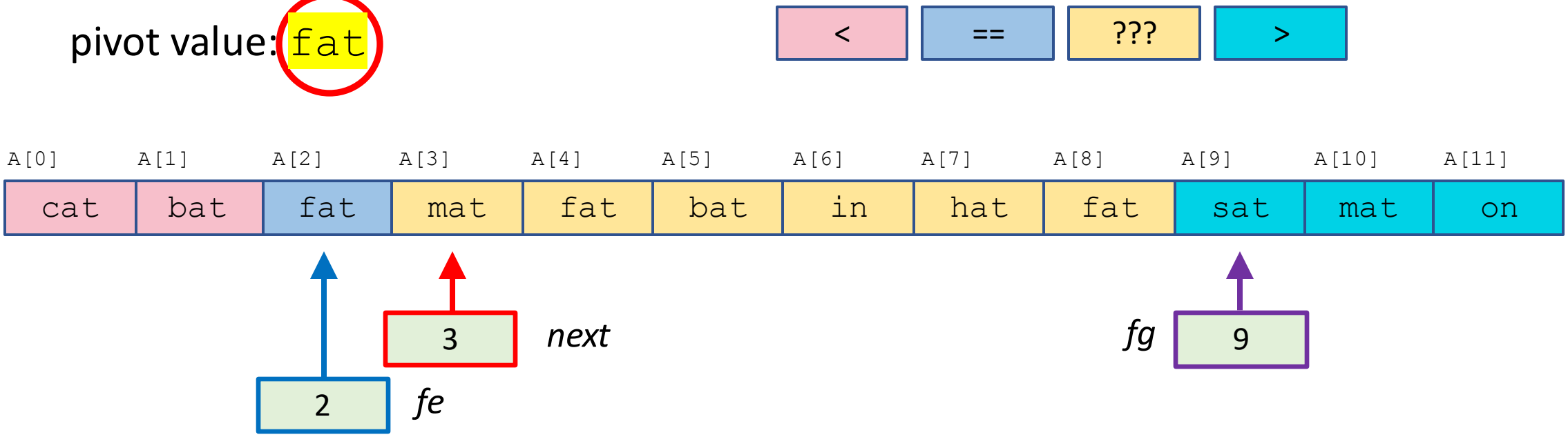

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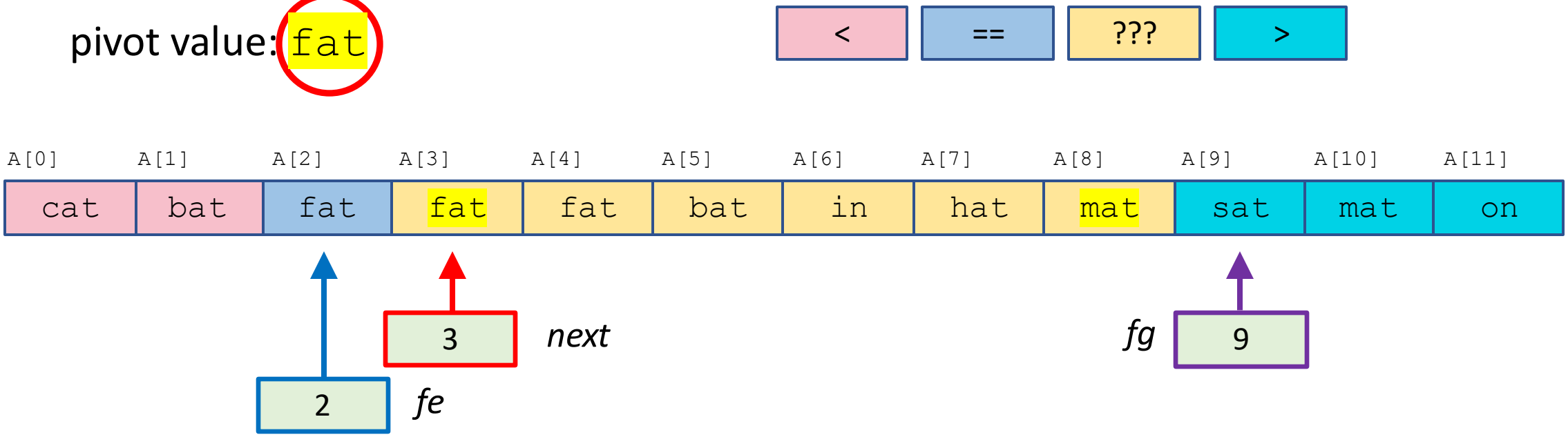

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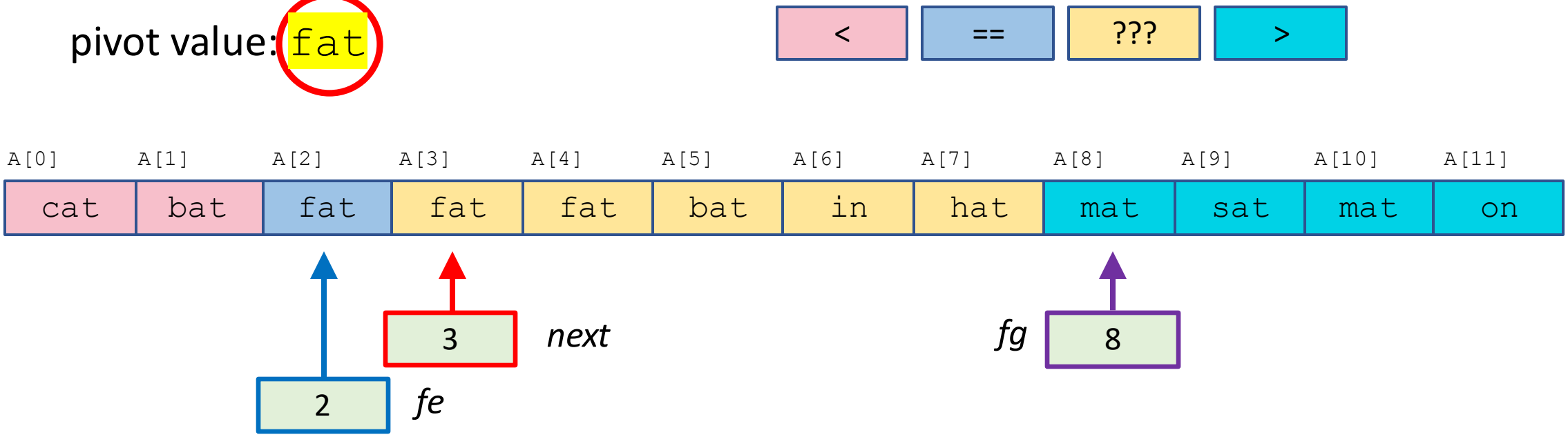

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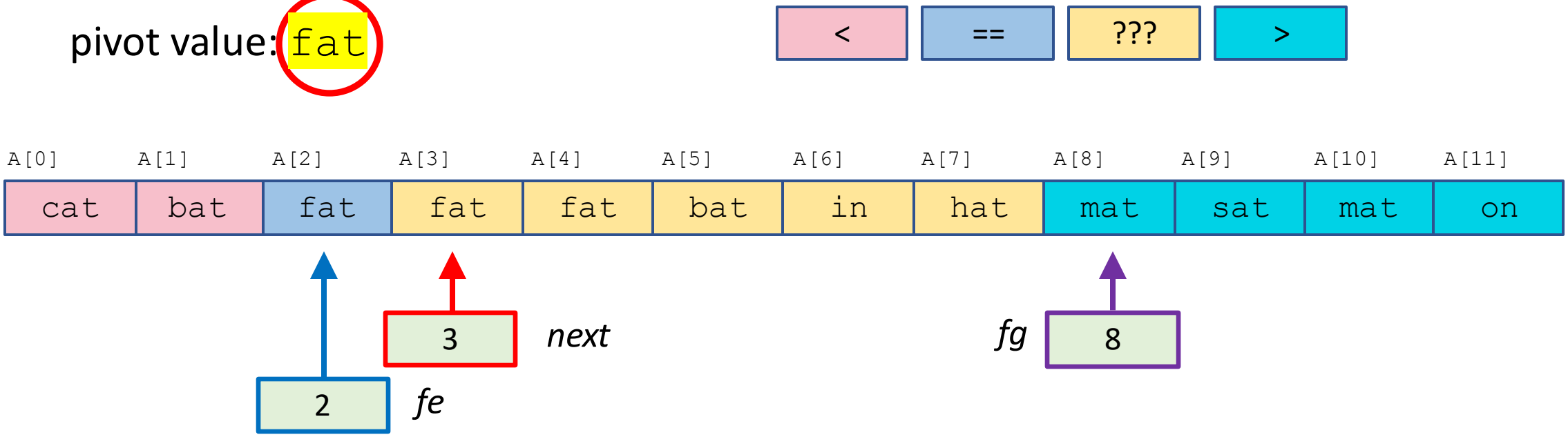

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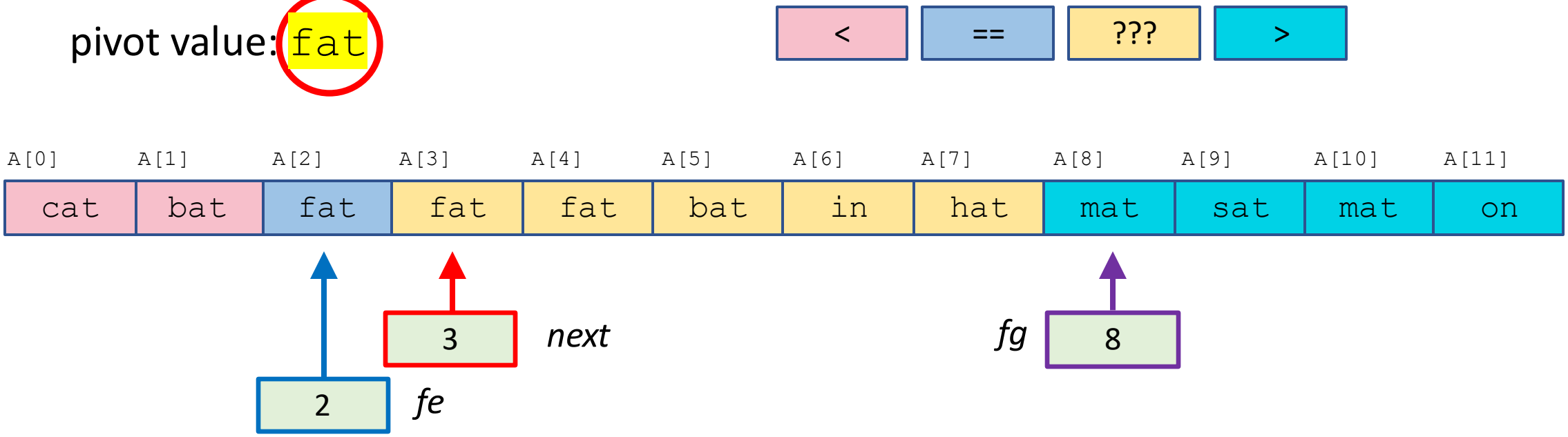


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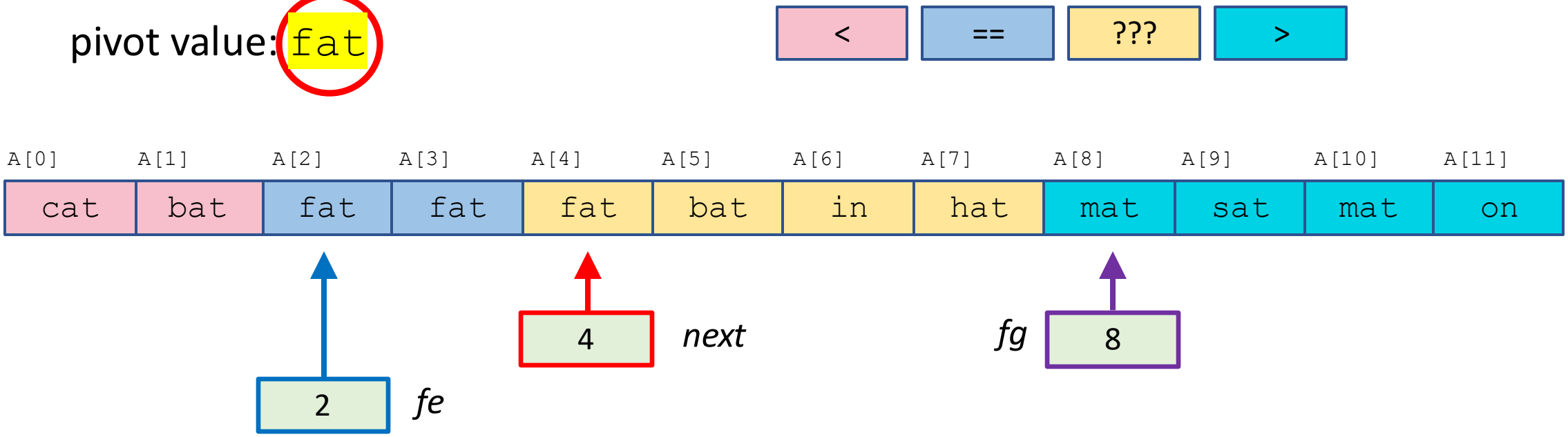


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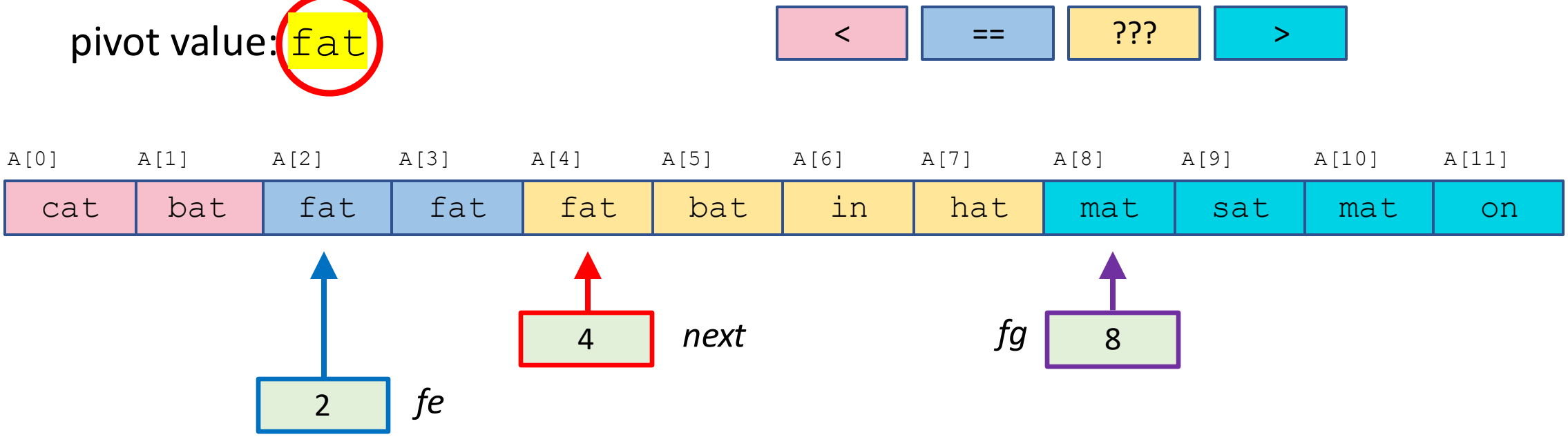


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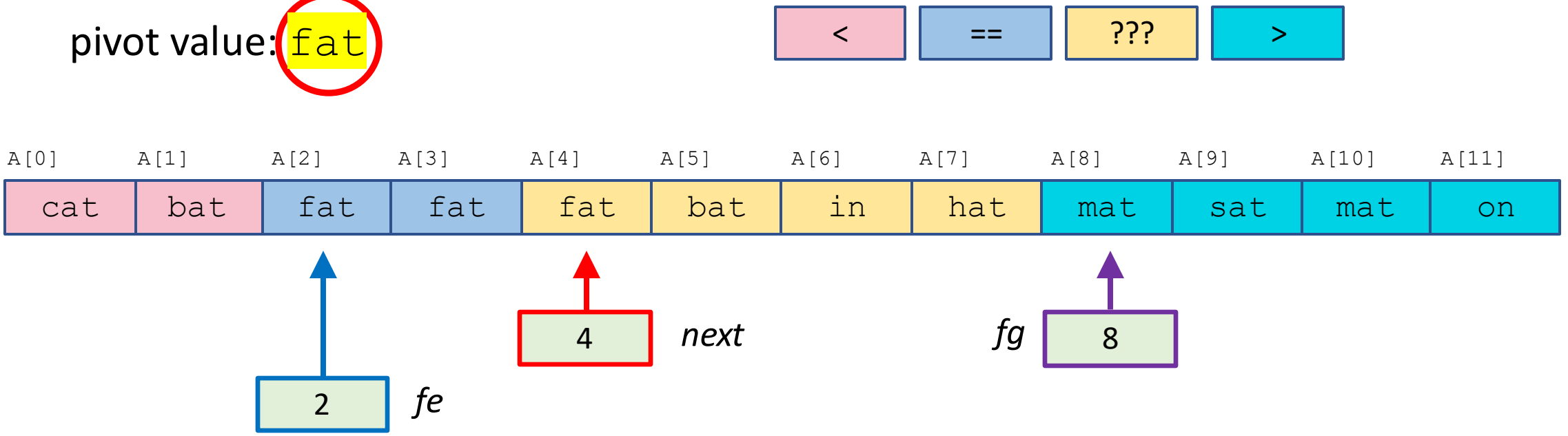


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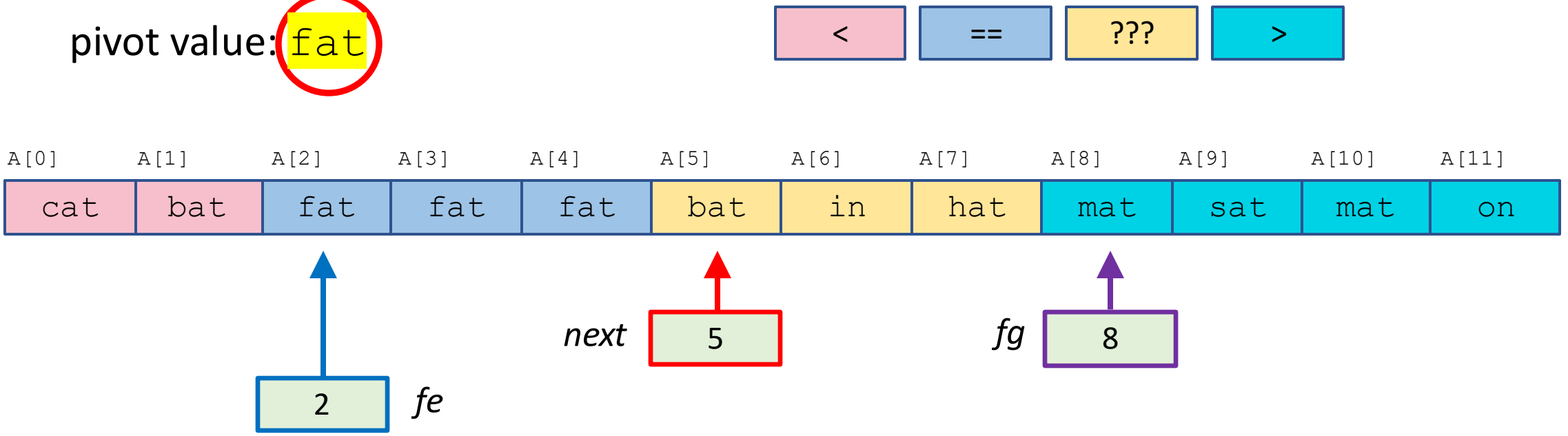


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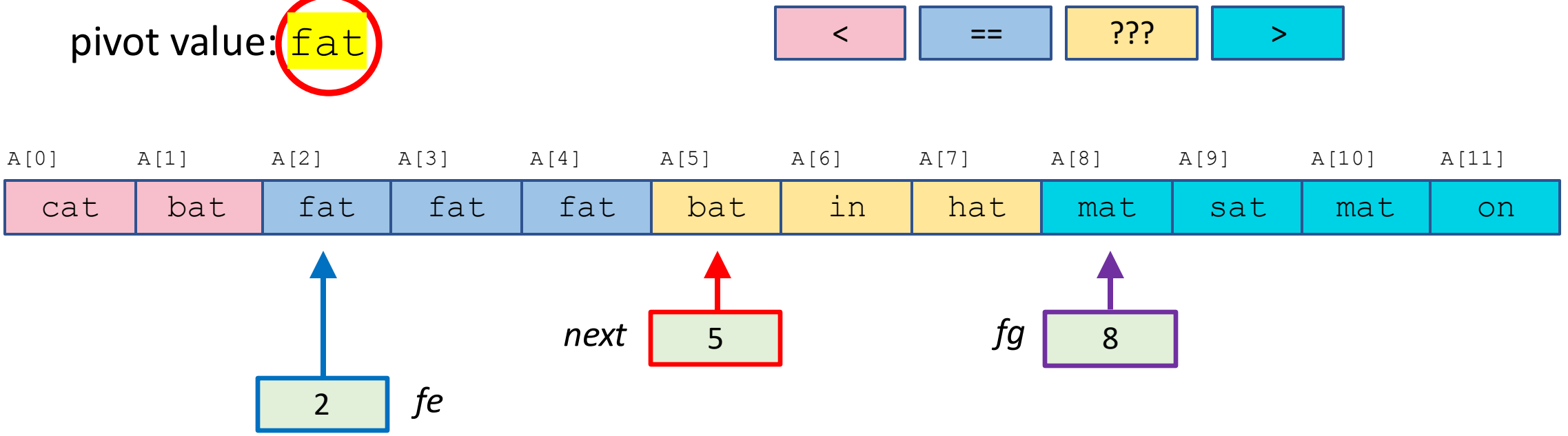

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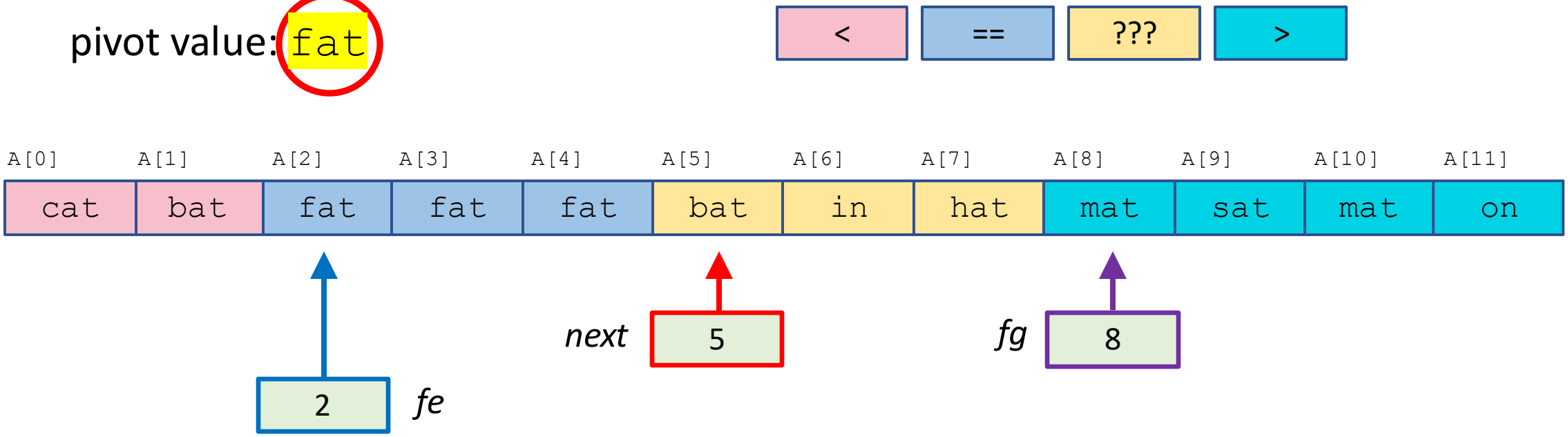

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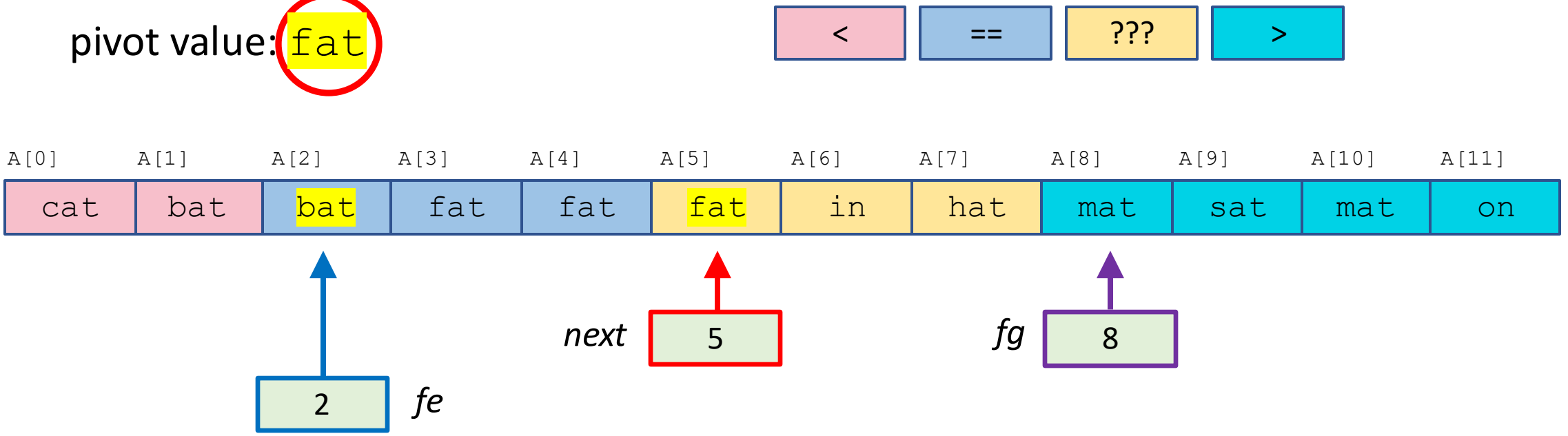

< == ??? >

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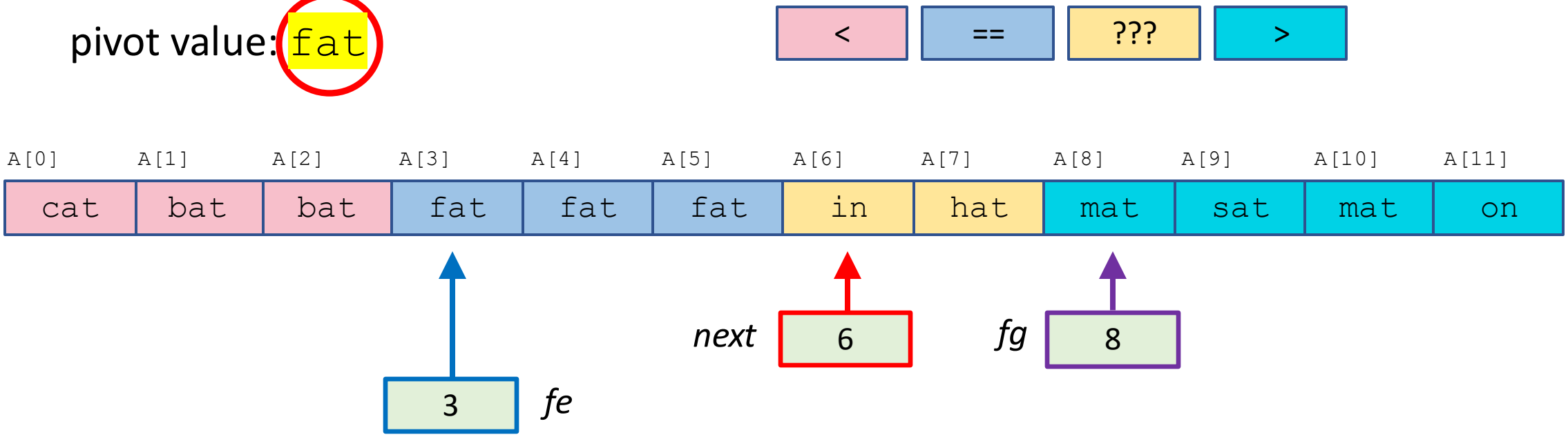

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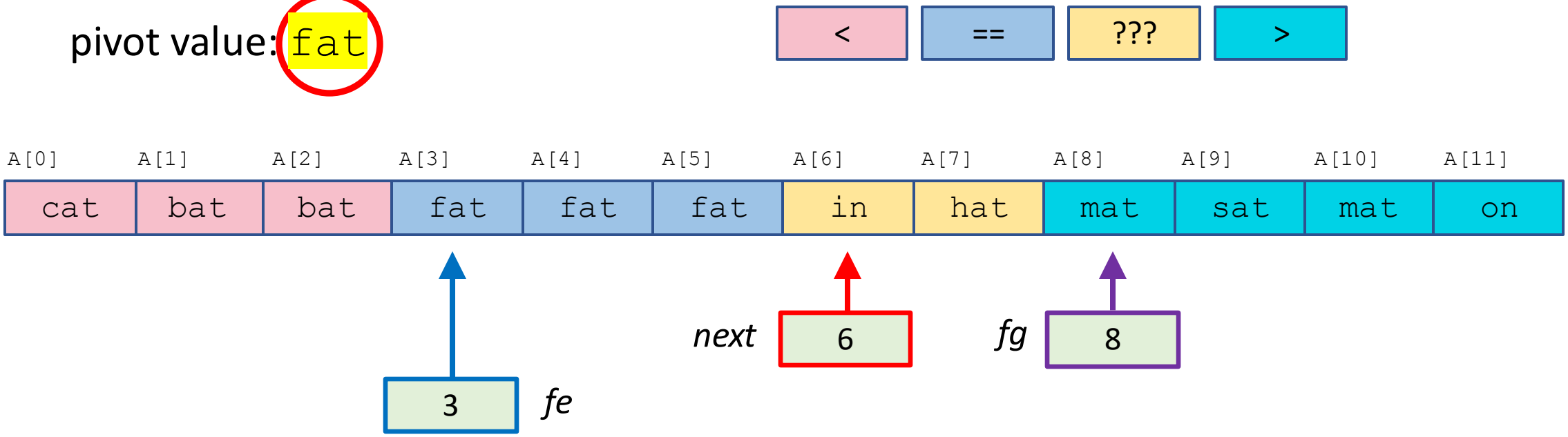
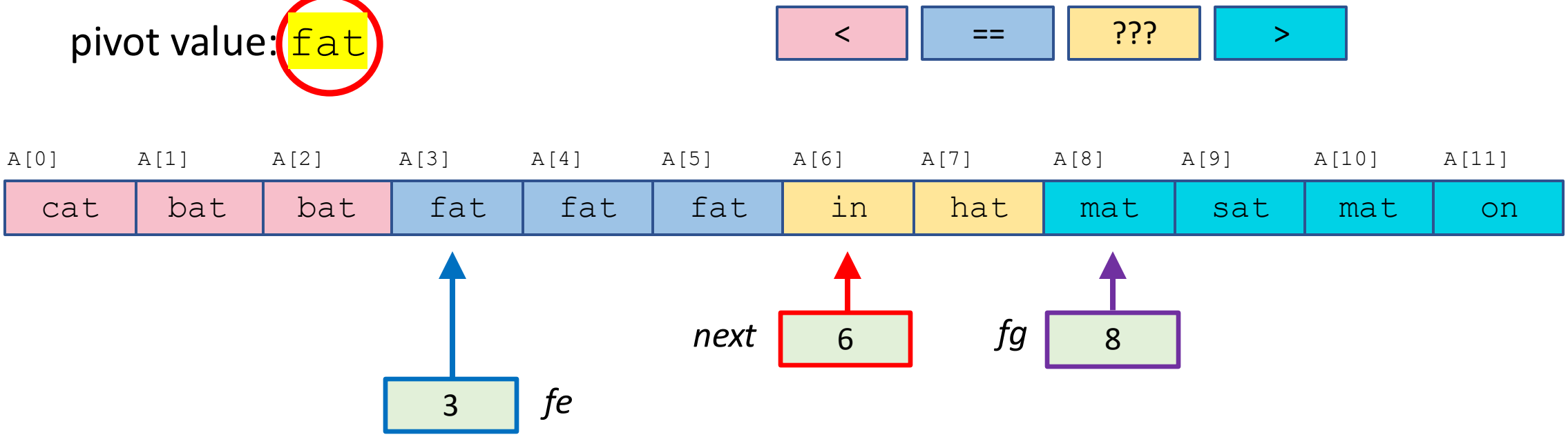
    next ← *next* + 1

pivot value: `fat`



next, *fe*, *fg* ← 0, 0, *n*
while *next* < *fg* **do**
  if *A*[*next*] < *p* **then**
    swap *A*[*fe*] and *A*[*next*]
    fe, *next* ← *fe* + 1, *next* + 1
  else if *A*[*next*] > *p* **then**
    swap *A*[*next*] and *A*[*fg* − 1]
    fg ← *fg* − 1
  else
    next ← *next* + 1

pivot value: `fat`



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  if *A*[*next*] < *p* **then**
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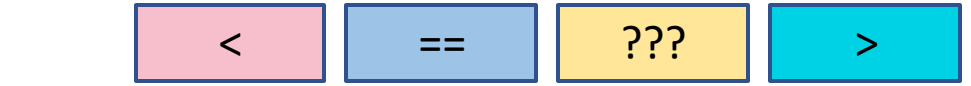
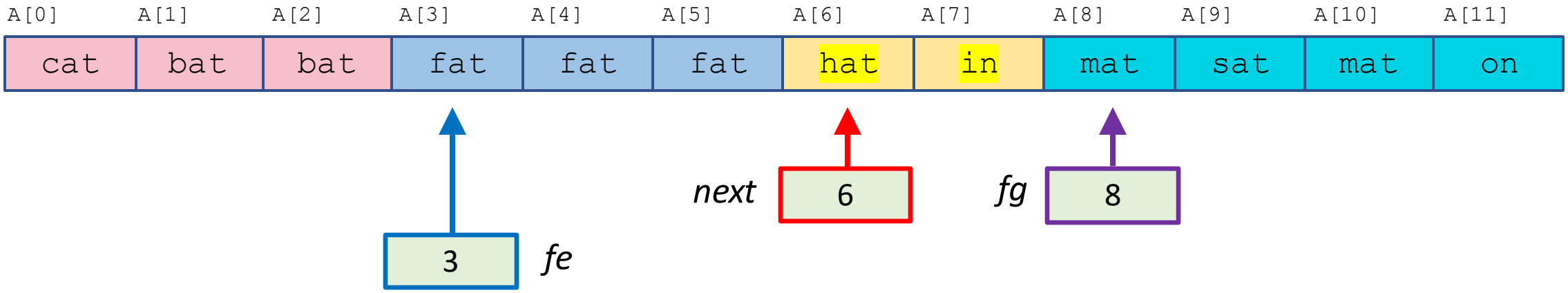
  else if *A*[*next*] > *p* **then**
    swap *A*[*next*] and *A*[*fg* − 1]
    fg ← *fg* − 1
  else
    next ← *next* + 1

pivot value: fat





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while *next* < *fg* **do**
  if *A*[*next*] < *p* **then**
    swap *A*[*fe*] and *A*[*next*]
    fe, *next* ← *fe* + 1, *next* + 1
  else if *A*[*next*] > *p* **then**
    swap *A*[*next*] and *A*[*fg* − 1]
    fg ← *fg* − 1
  else
    next ← *next* + 1

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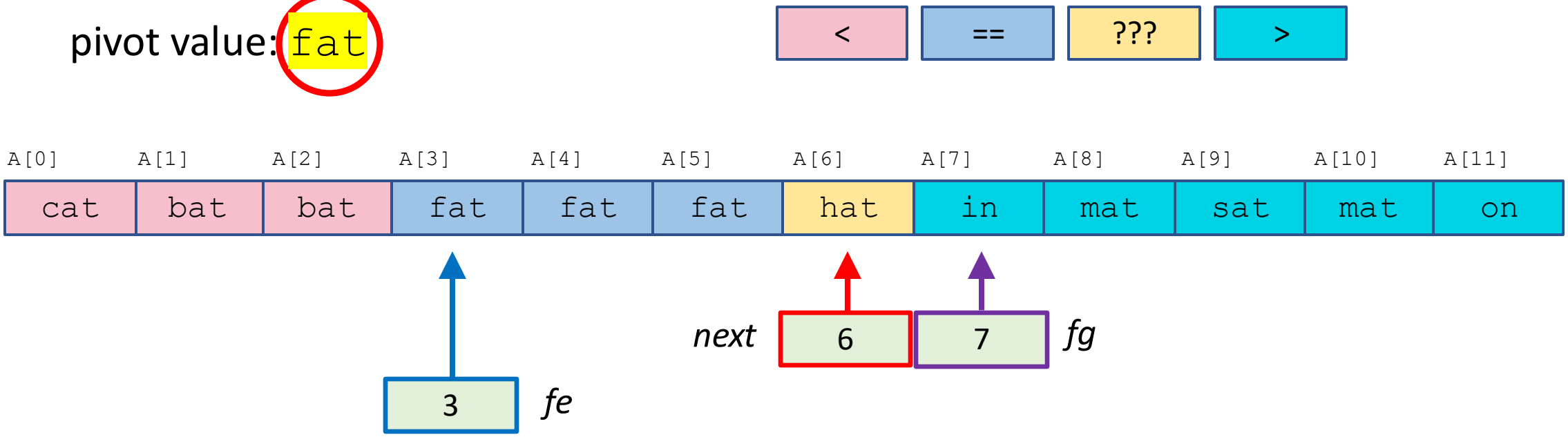
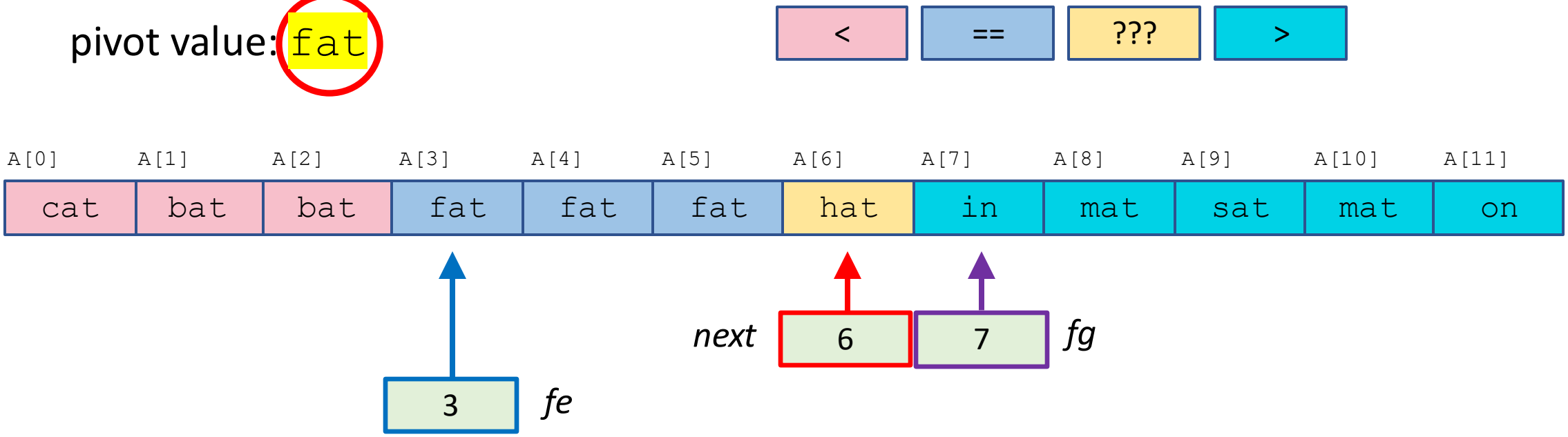



```
next, fe, fg ← 0, 0, n
while next < fg do
    if A[next] < p then
        swap A[fe] and A[next]
        fe, next ← fe + 1, next + 1
    else if A[next] > p then
        swap A[next] and A[fg − 1]
        fg ← fg − 1
    else
        next ← next + 1
```

pivot value: `fat`



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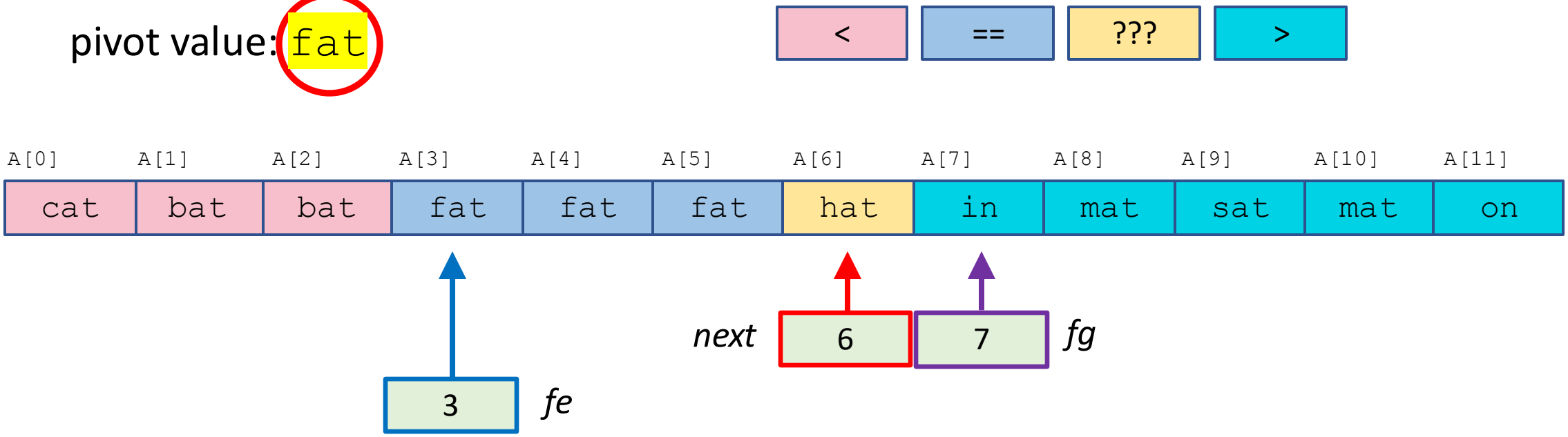
while *next* < *fg* **do**
  if *A*[*next*] < *p* **then**
    swap *A*[*fe*] and *A*[*next*]
    fe, *next* ← *fe* + 1, *next* + 1
  else if *A*[*next*] > *p* **then**
    swap *A*[*next*] and *A*[*fg* − 1]
    fg ← *fg* − 1
  else
    next ← *next* + 1

pivot value: `fat`

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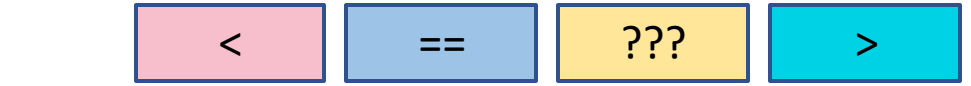
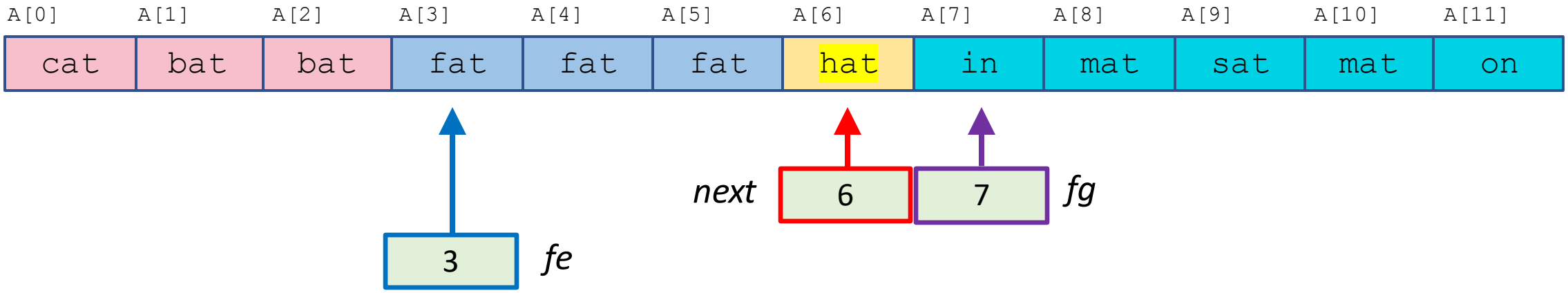
next, fe, fg ← 0, 0, n
while next < fg do
    if A[next] < p then
        swap A[fe] and A[next]
        fe, next ← fe + 1, next + 1
    else if A[next] > p then
        swap A[next] and A[fg − 1]
        fg ← fg − 1
    else
        next ← next + 1
```

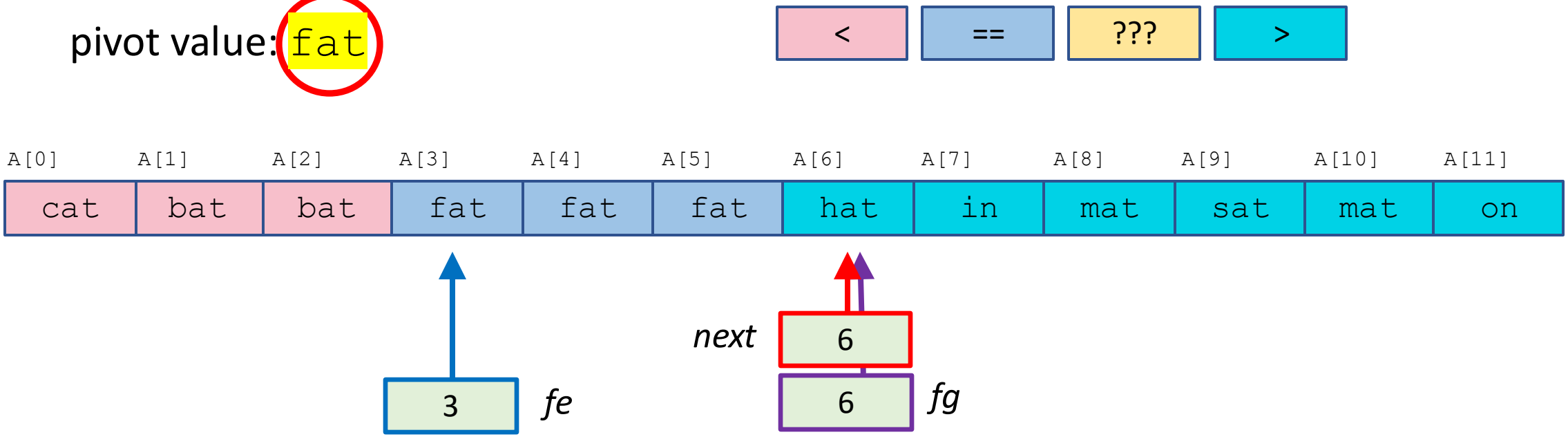
pivot value: fat





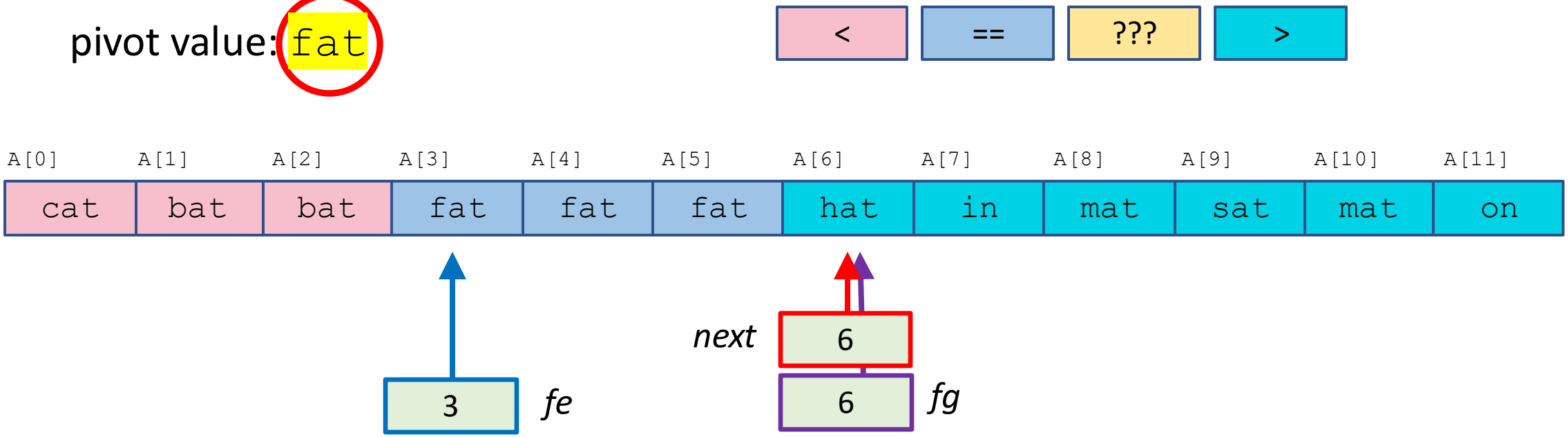
next, *fe*, *fg* ← 0, 0, *n*
while *next* < *fg* **do**
  if *A*[*next*] < *p* **then**
    swap *A*[*fe*] and *A*[*next*]
    fe, *next* ← *fe* + 1, *next* + 1
  else if *A*[*next*] > *p* **then**
    swap *A*[*next*] and *A*[*fg* − 1]
    fg ← *fg* − 1
  else
    next ← *next* + 1

pivot value: `fat`



next, *fe*, *fg* ← 0, 0, *n*
while *next* < *fg* **do**
  if *A*[*next*] < *p* **then**
    swap *A*[*fe*] and *A*[*next*]
    fe, *next* ← *fe* + 1, *next* + 1
  else if *A*[*next*] > *p* **then**
    swap *A*[*next*] and *A*[*fg* − 1]
    fg ← *fg* − 1
  else
    next ← *next* + 1

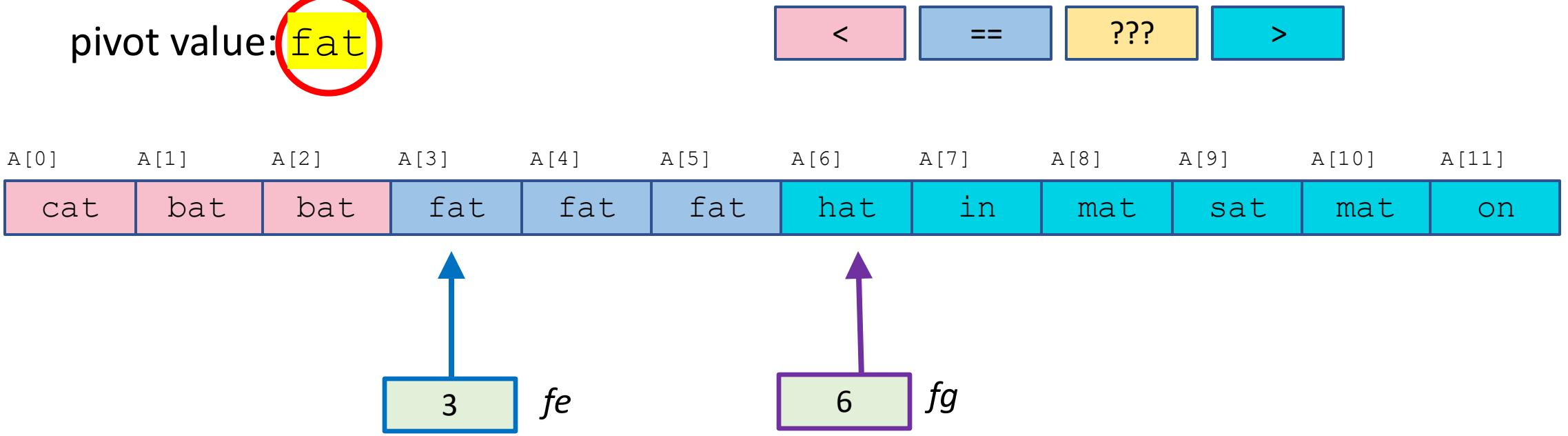
pivot value: `fat`



< == ??? >

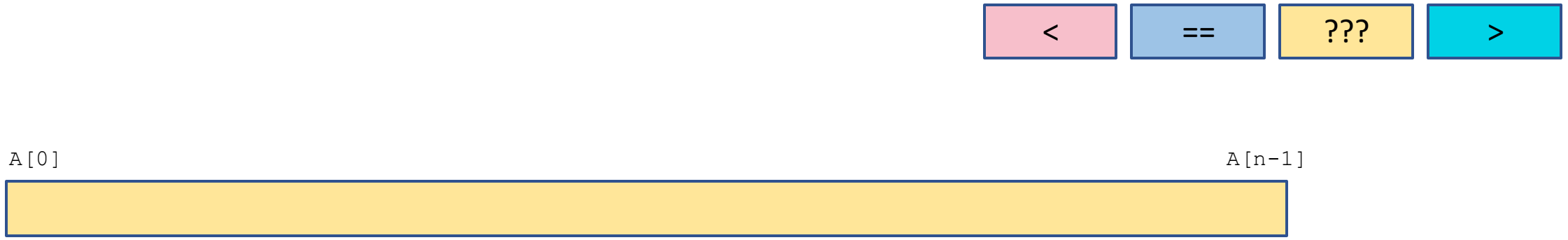
next, *fe*, *fg* ← 0, 0, *n*
while *next* < *fg* **do**
  if *A*[*next*] < *p* **then**
    swap *A*[*fe*] and *A*[*next*]
    fe, *next* ← *fe* + 1, *next* + 1
  else if *A*[*next*] > *p* **then**
    swap *A*[*next*] and *A*[*fg* − 1]
    fg ← *fg* − 1
  else
    next ← *next* + 1

pivot value: fat

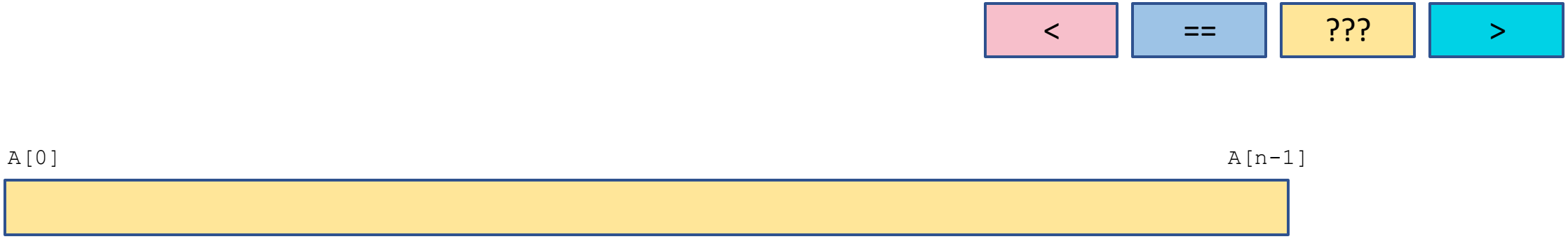


three-way partition is DONE...

can now go back to the big picture



quicksort(*A*, *n*):
  if *n* > 1 **then**
    p ← any element in *A*[0 .. *n* − 1]
    (*fe*, *fg*) ← partition(*A*, *n*, *p*)
    quicksort(*A*, *fe*)
    quicksort(*A* + *fg*, *n* − *fg*)
  return



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    quicksort(*A* + *fg*, *n* − *fg*)
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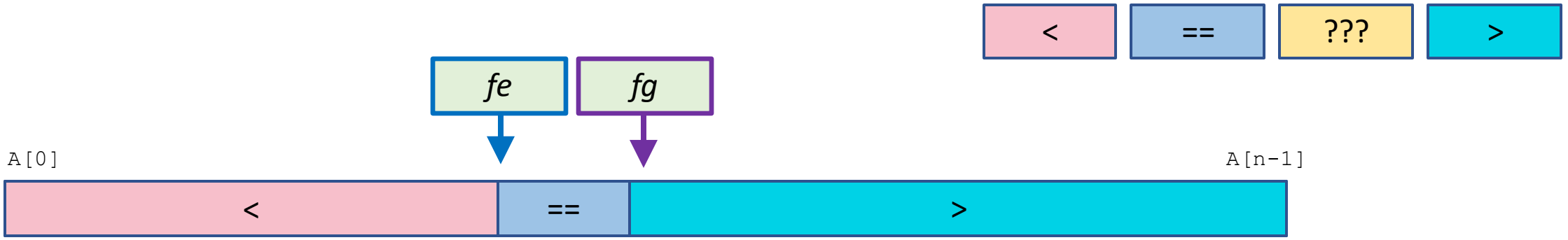
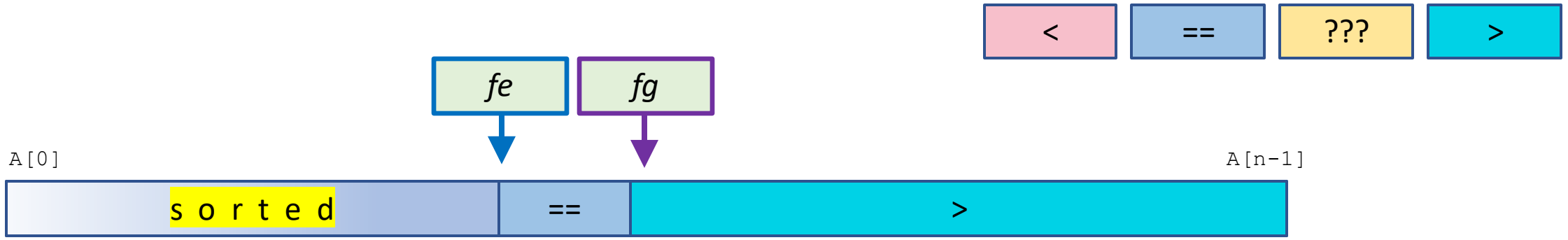
    quicksort(*A*, *fe*)
    quicksort(*A* + *fg*, *n* − *fg*)
  return




```
quicksort(A, n):
    if n > 1 then
        p ← any element in A[0 .. n − 1]
        (fe, fg) ← partition(A, n, p)
        quicksort(A, fe)
        quicksort(A + fg, n − fg)
    return
```



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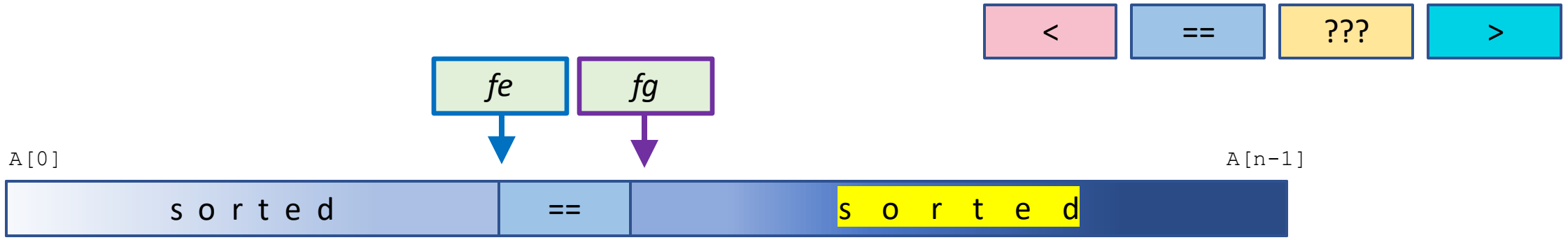
    quicksort(*A*, *fe*)
    quicksort(*A* + *fg*, *n* − *fg*)
  return



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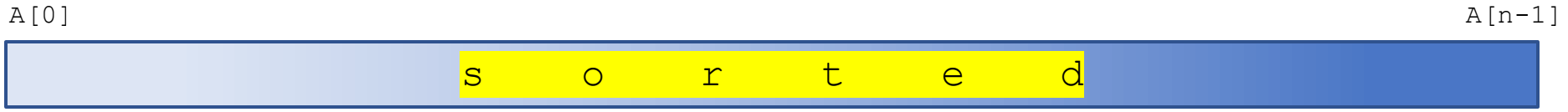
    quicksort(*A*, *fe*)
    quicksort(*A* + *fg*, *n* − *fg*)
  return

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quicksort(A, n):
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        (fe, fg) ← partition(A, n, p)
        quicksort(A, fe)
        quicksort(A + fg, n − fg)
    return
```
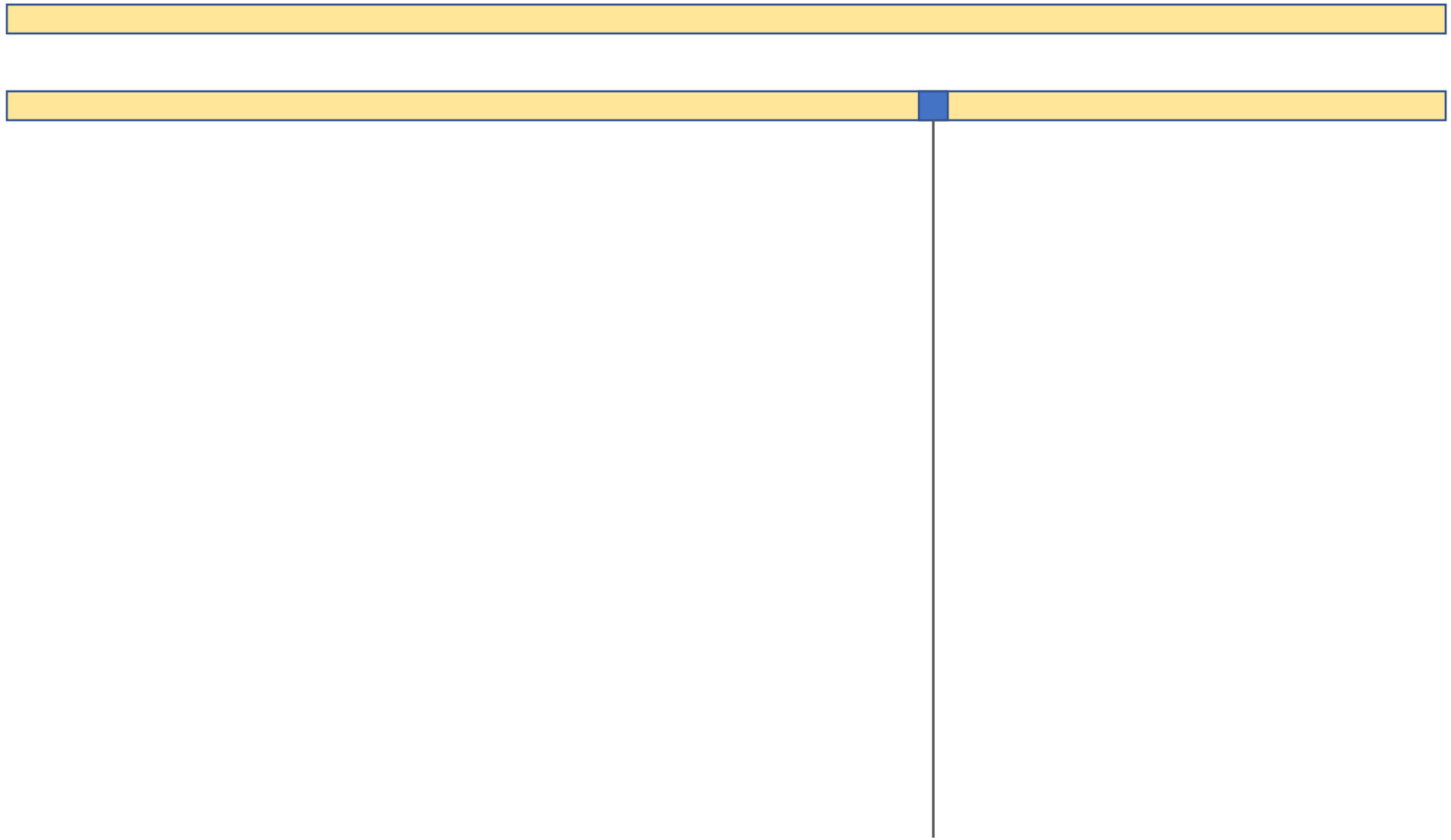
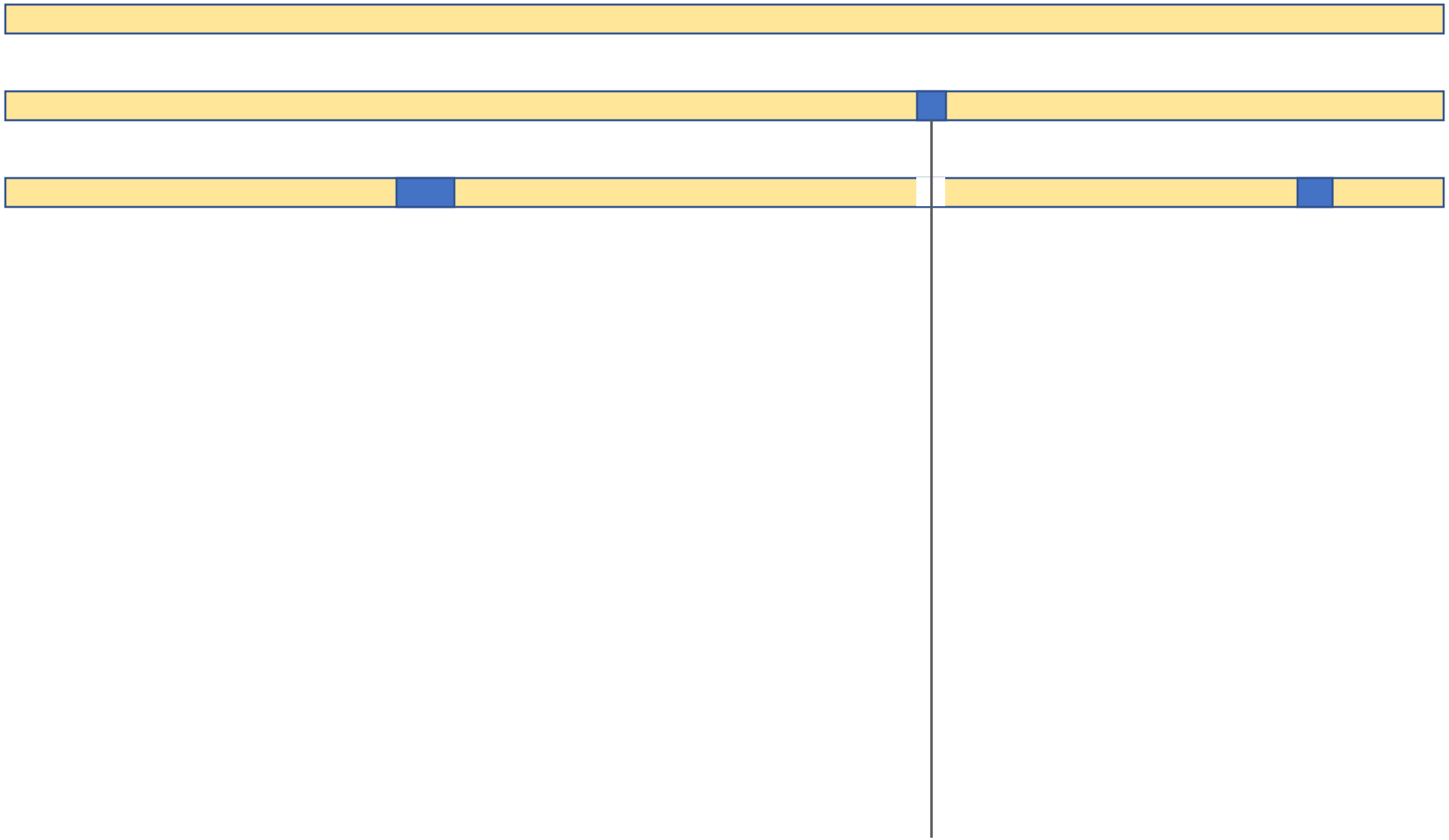
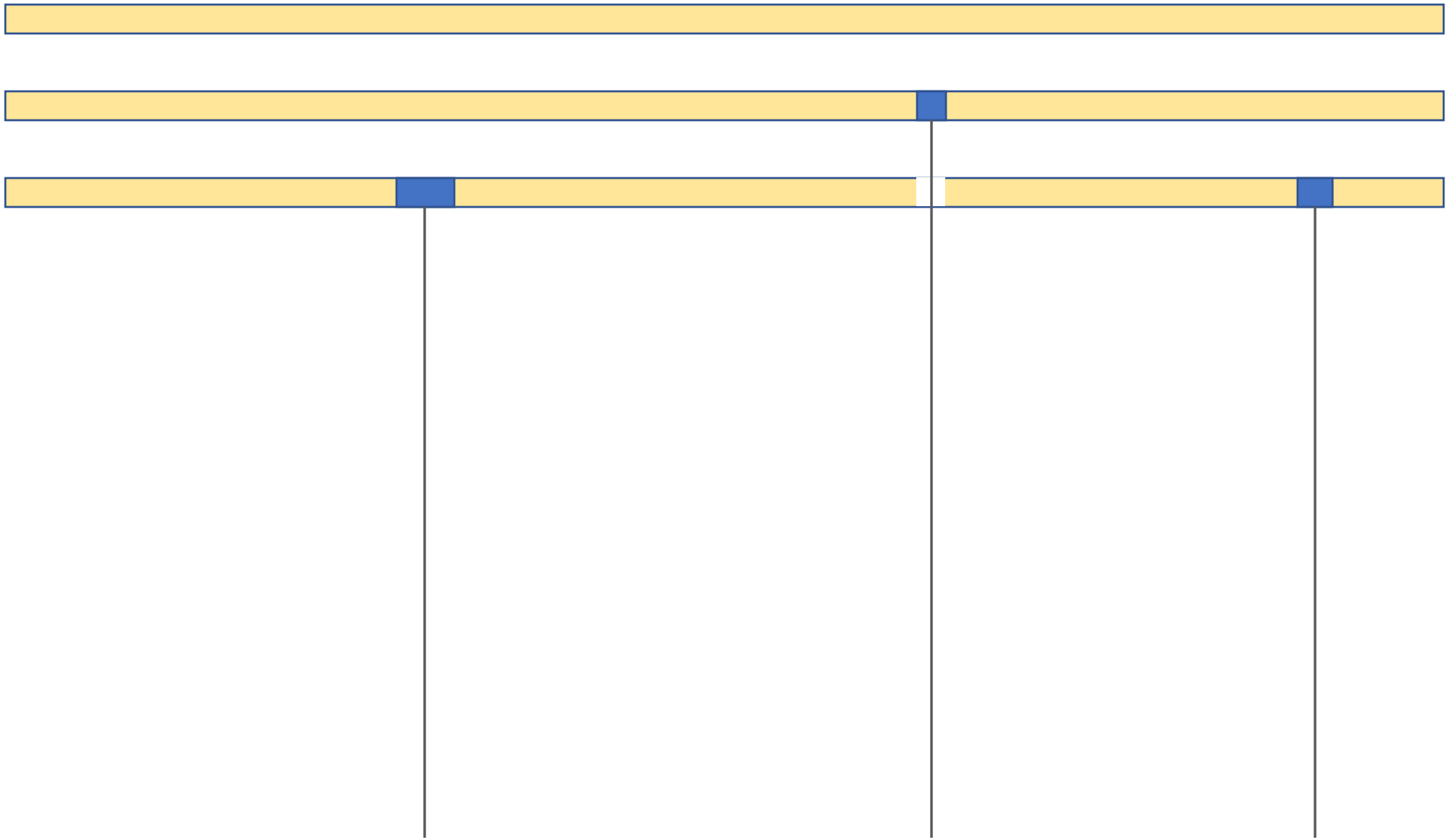
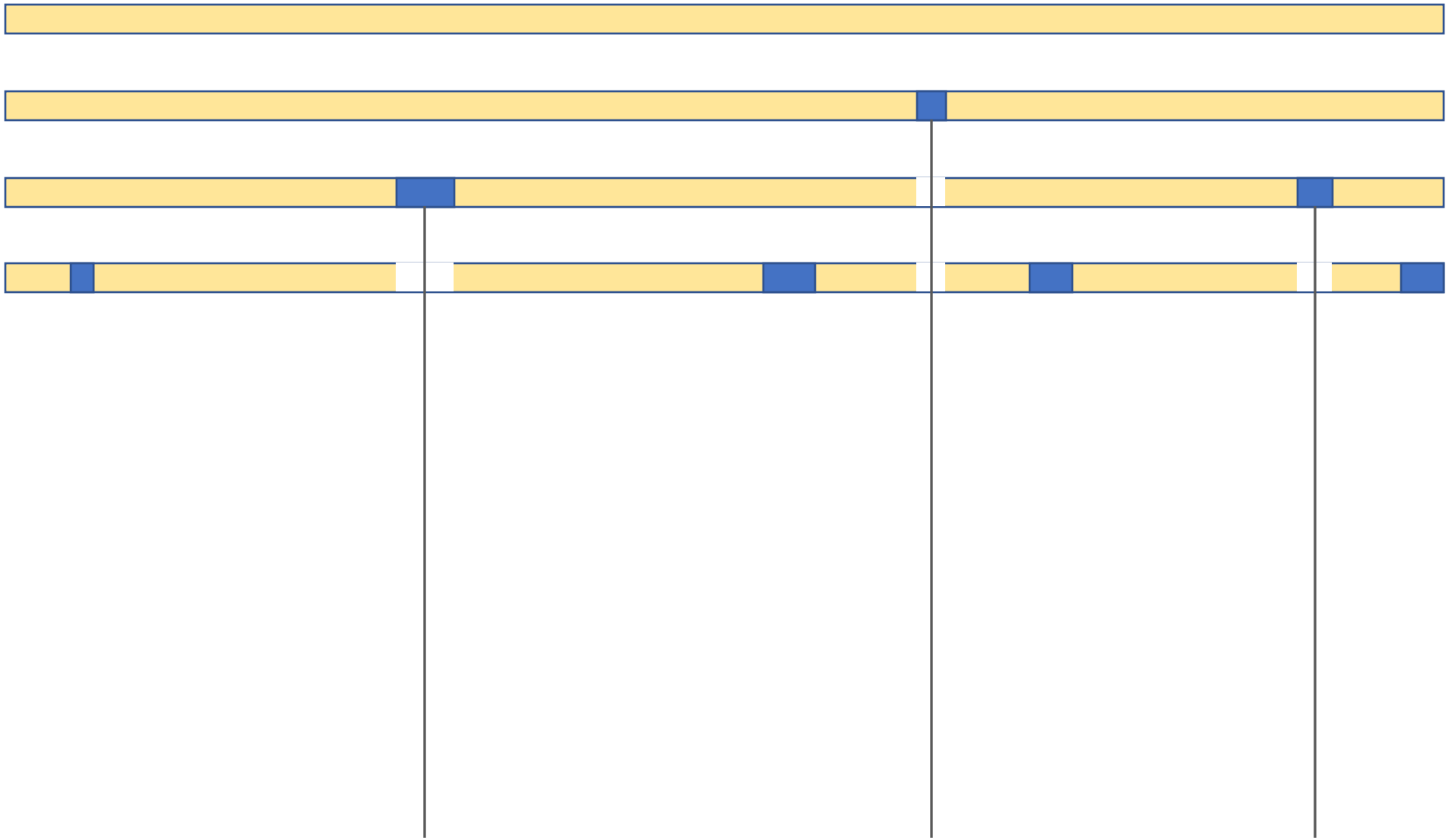
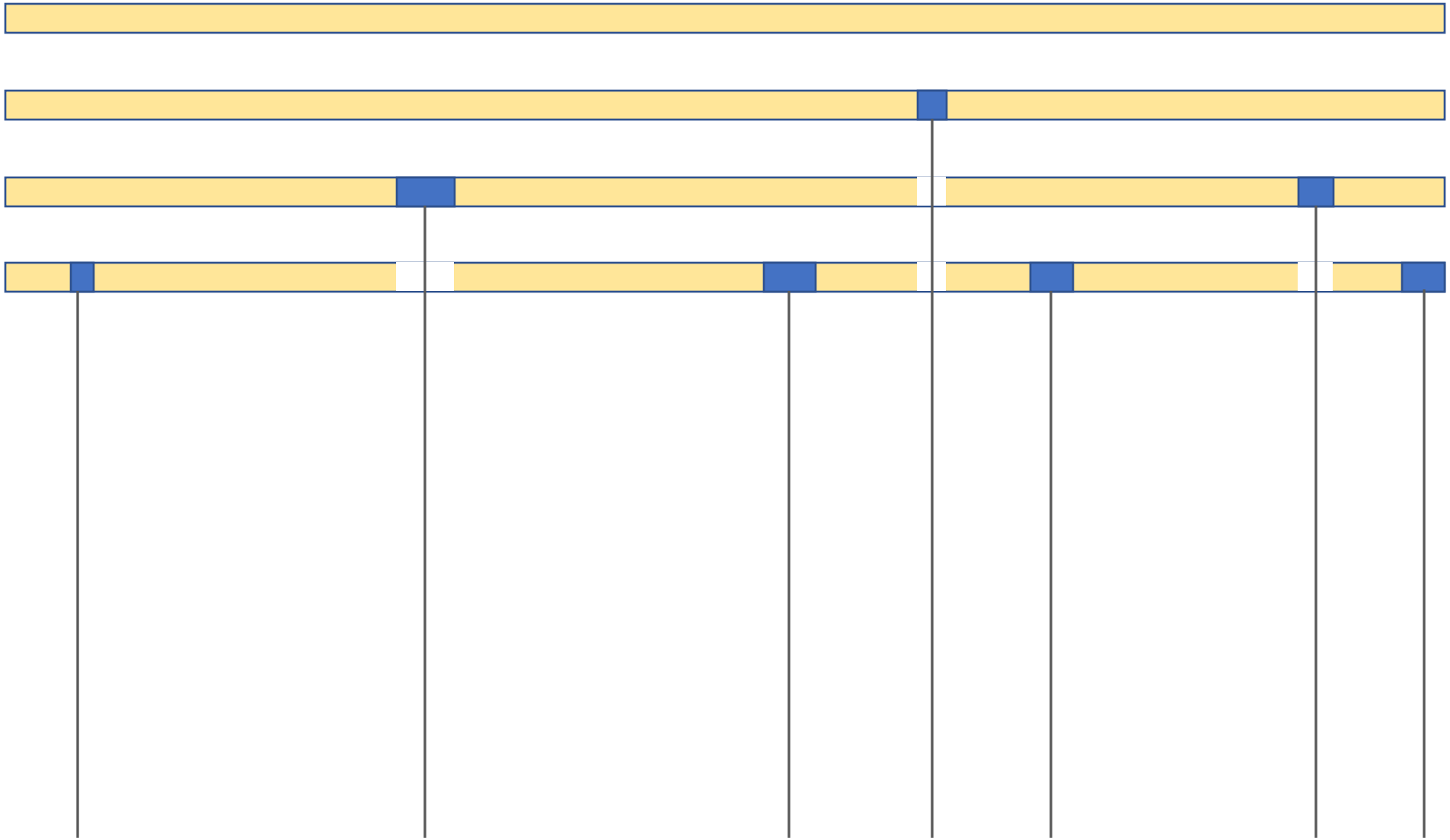

quicksort
execution time??

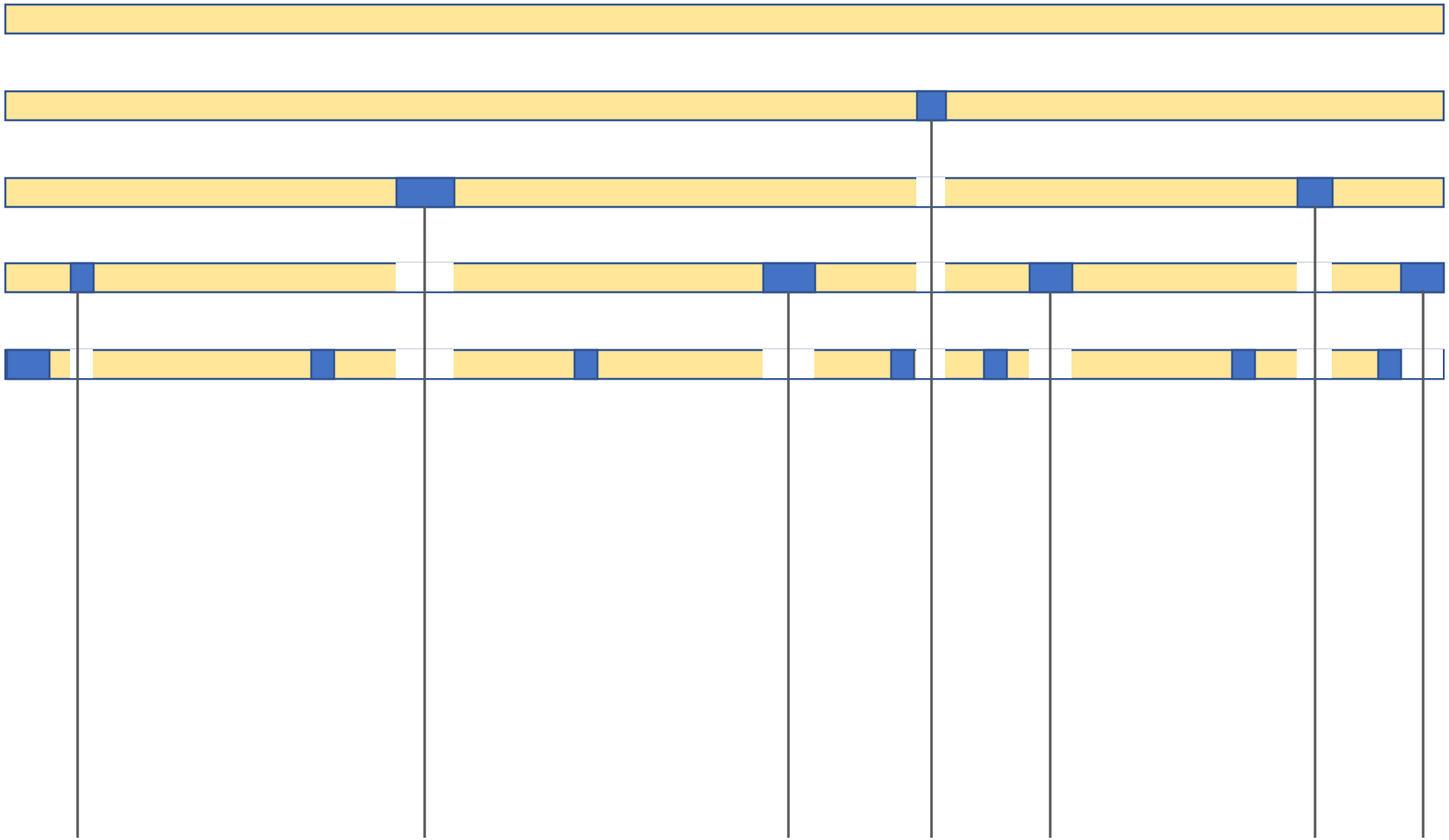


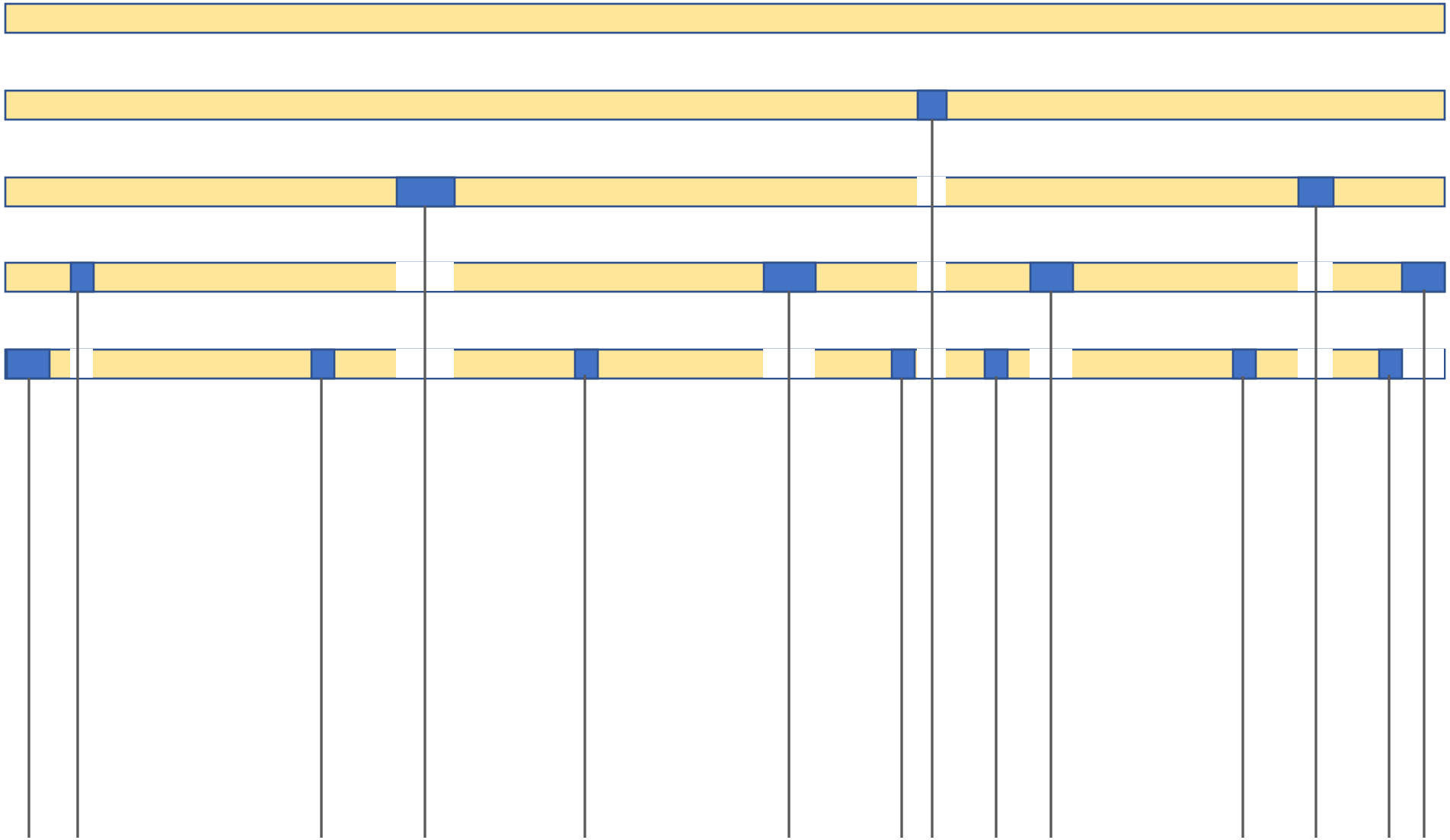


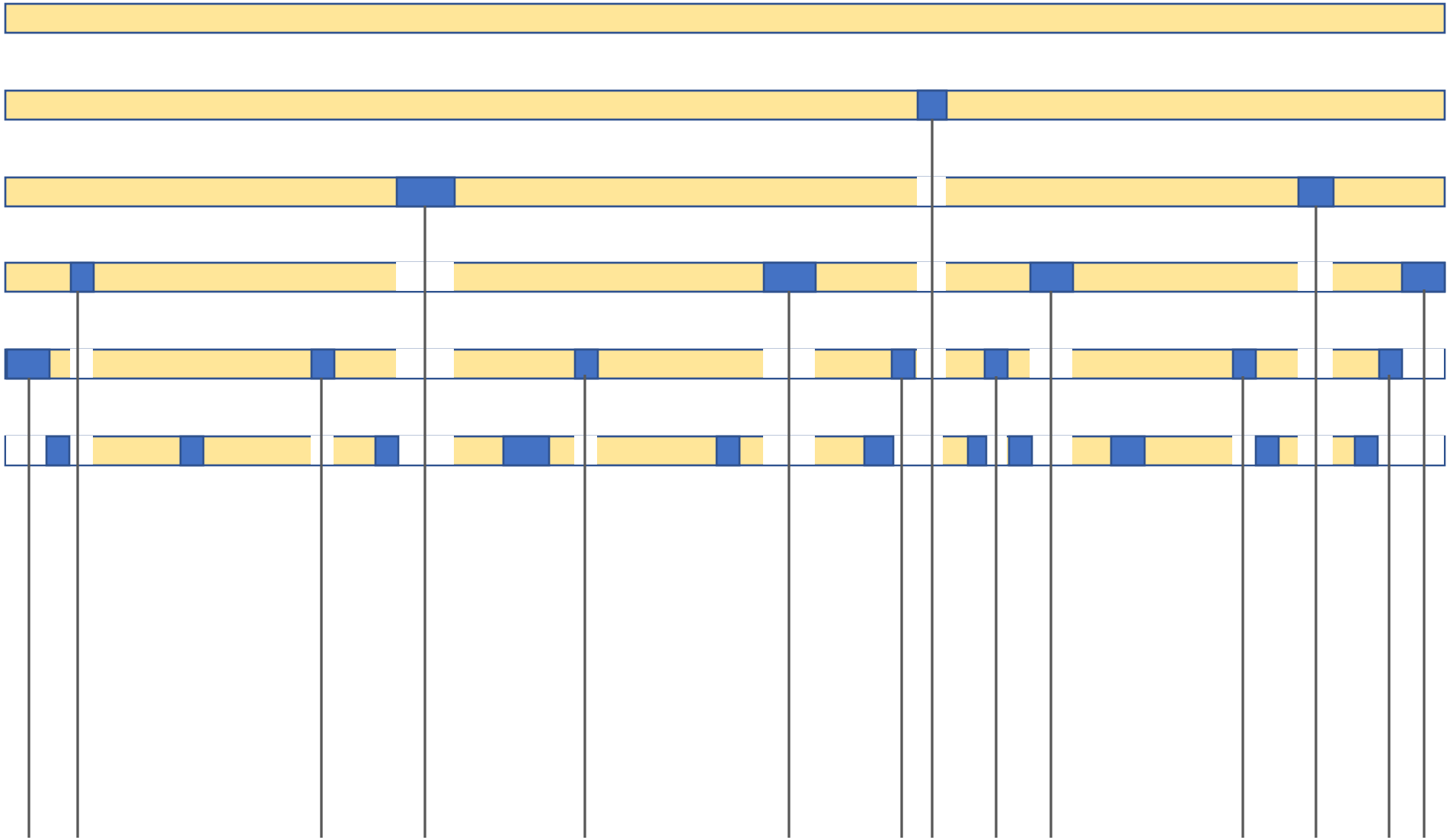


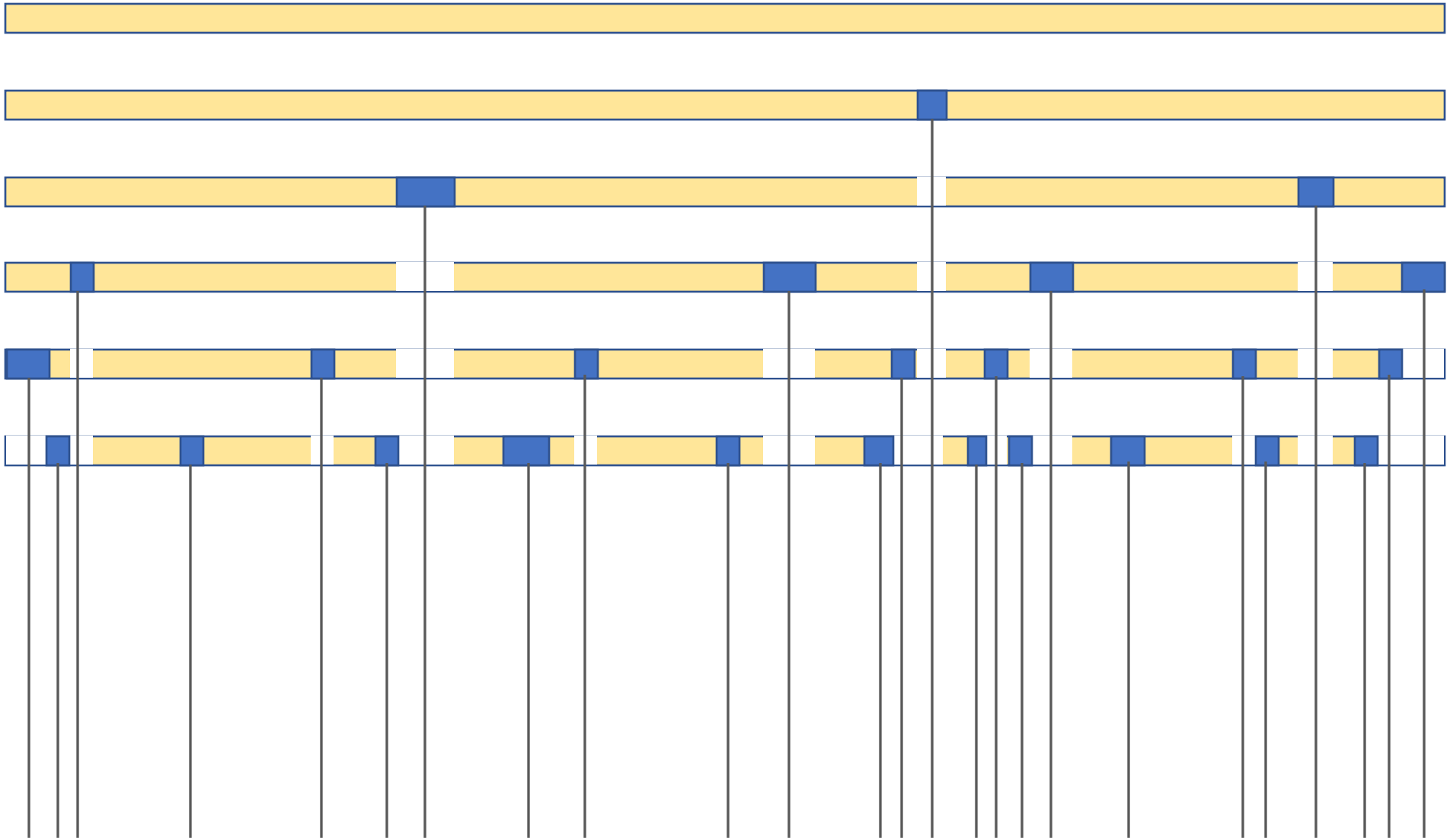


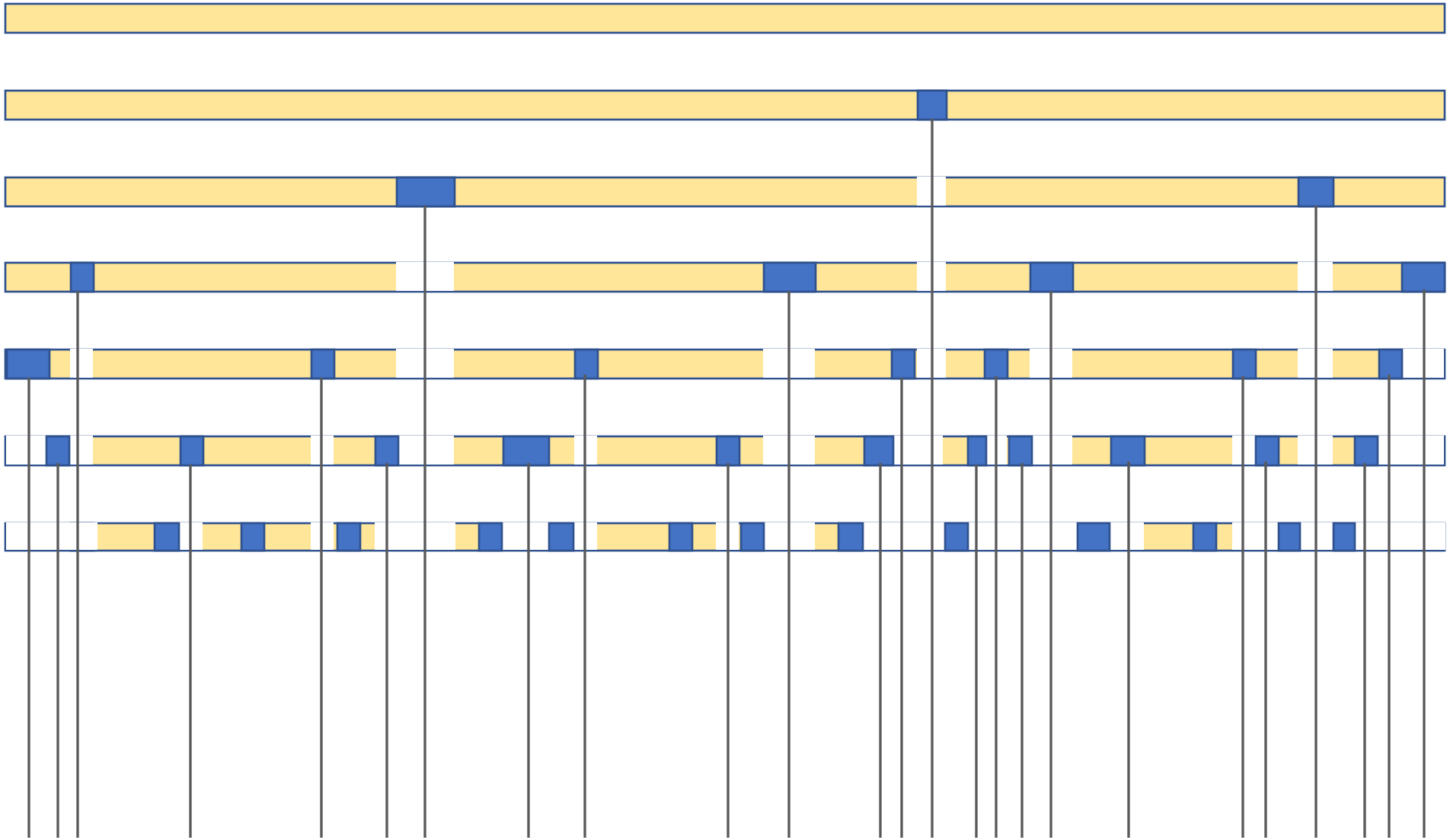


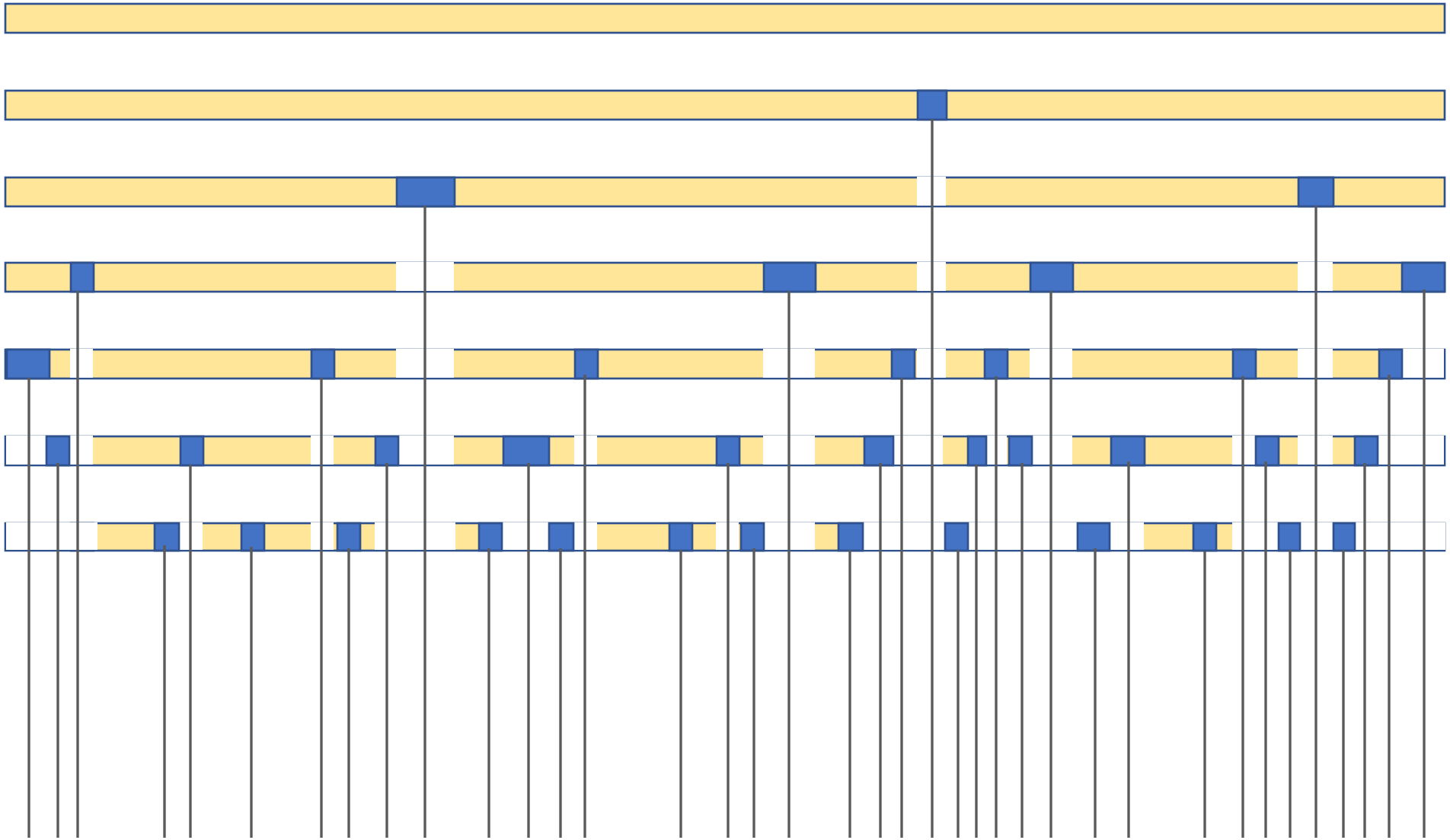


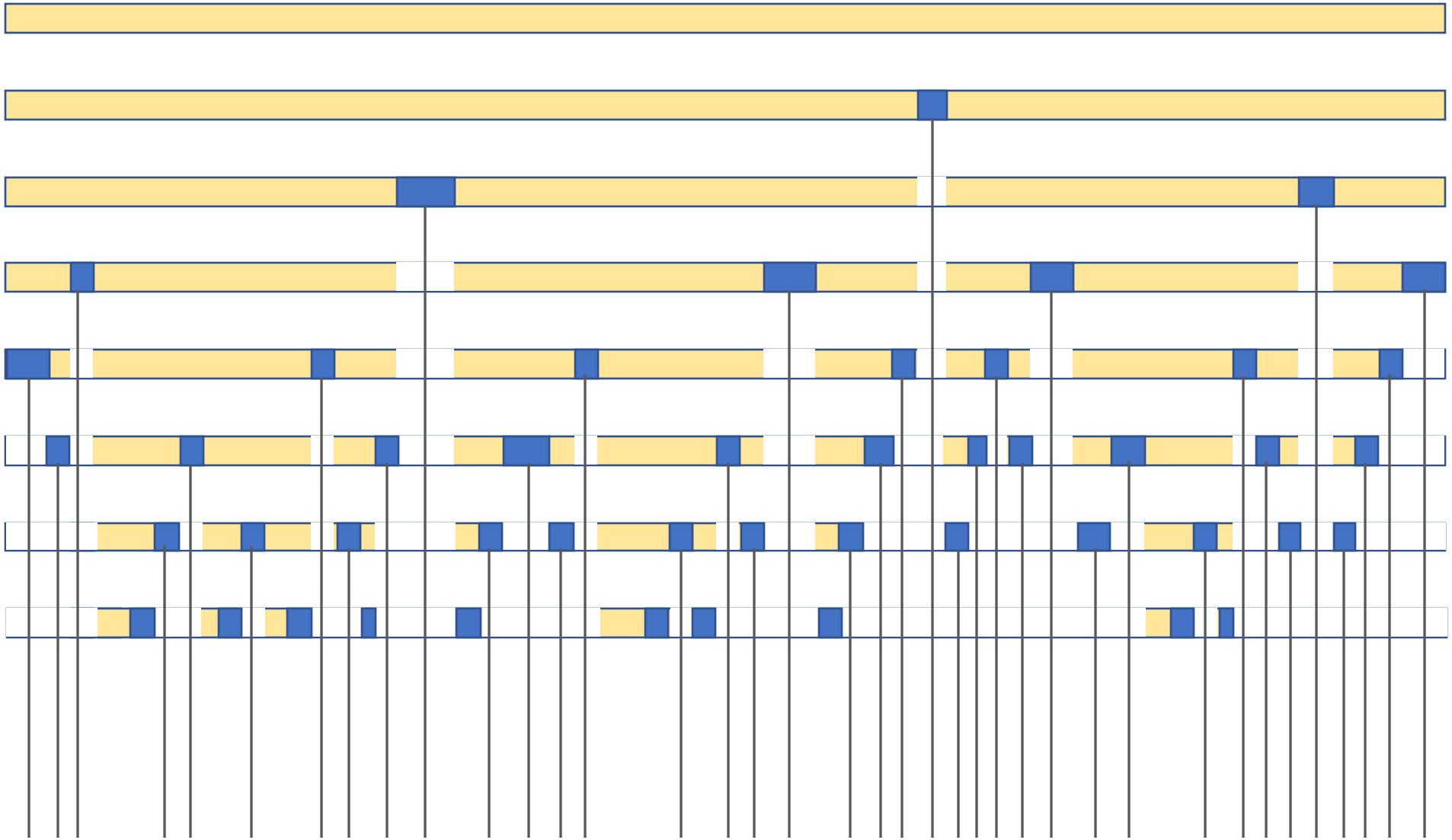


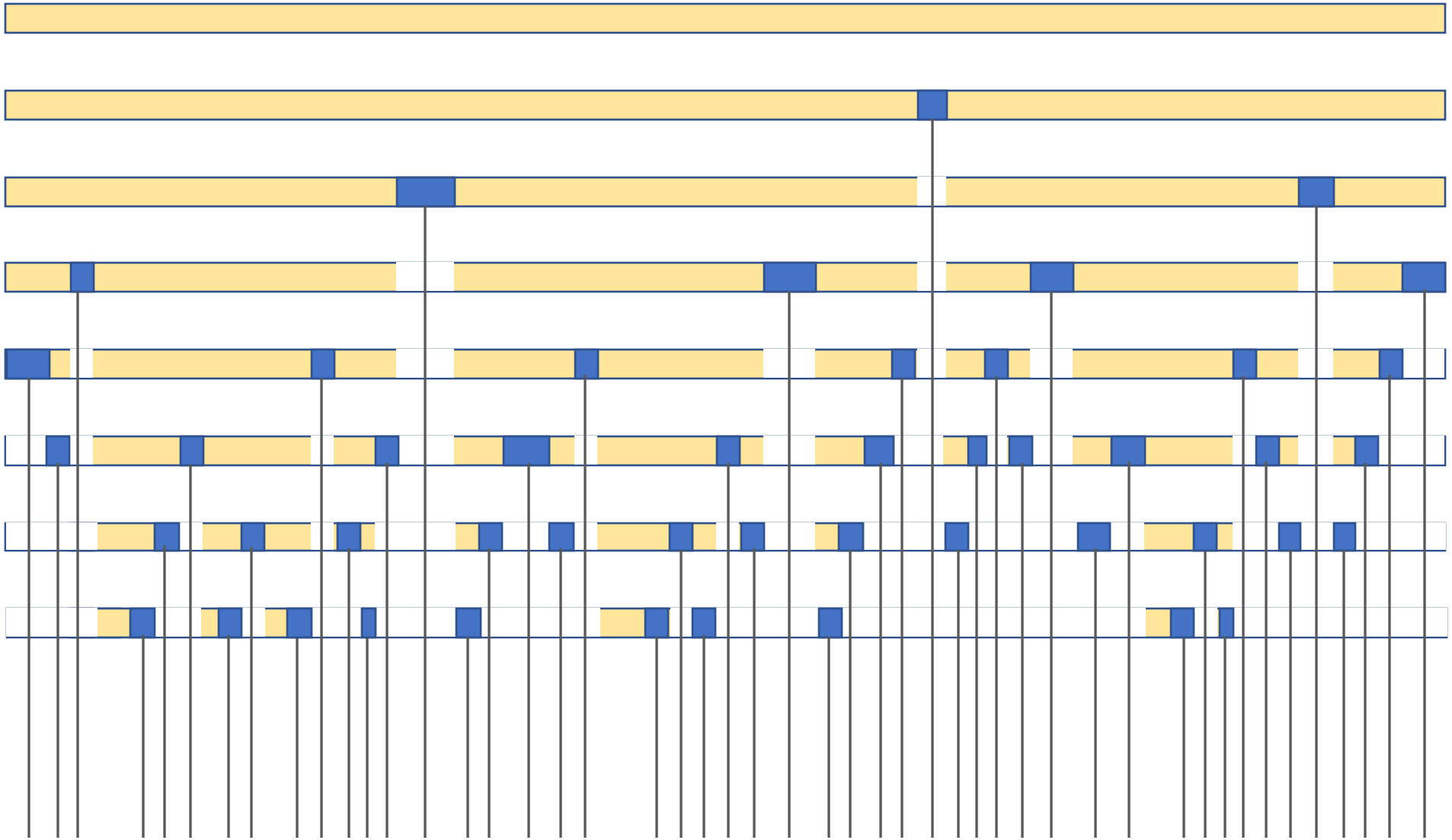


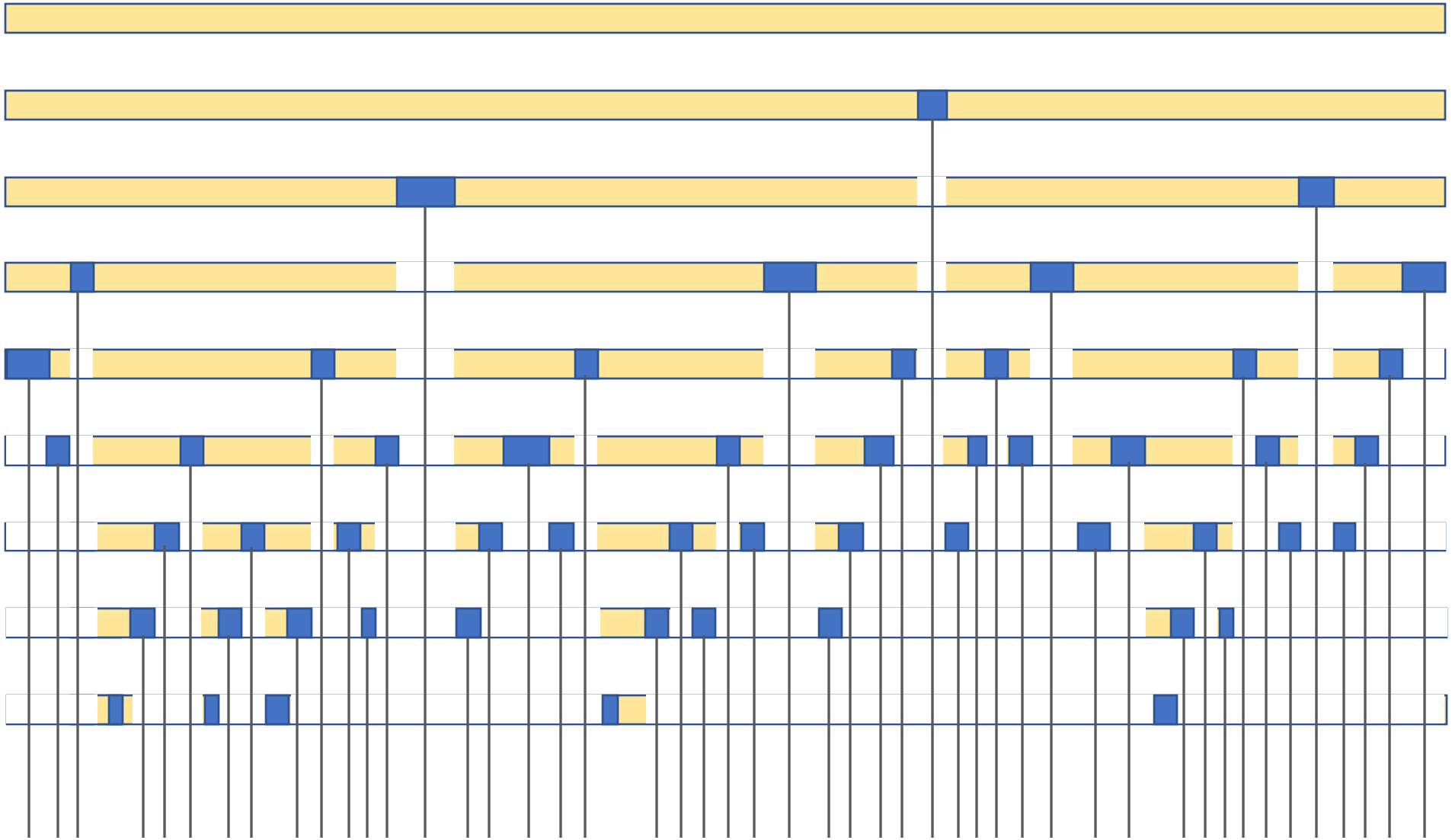


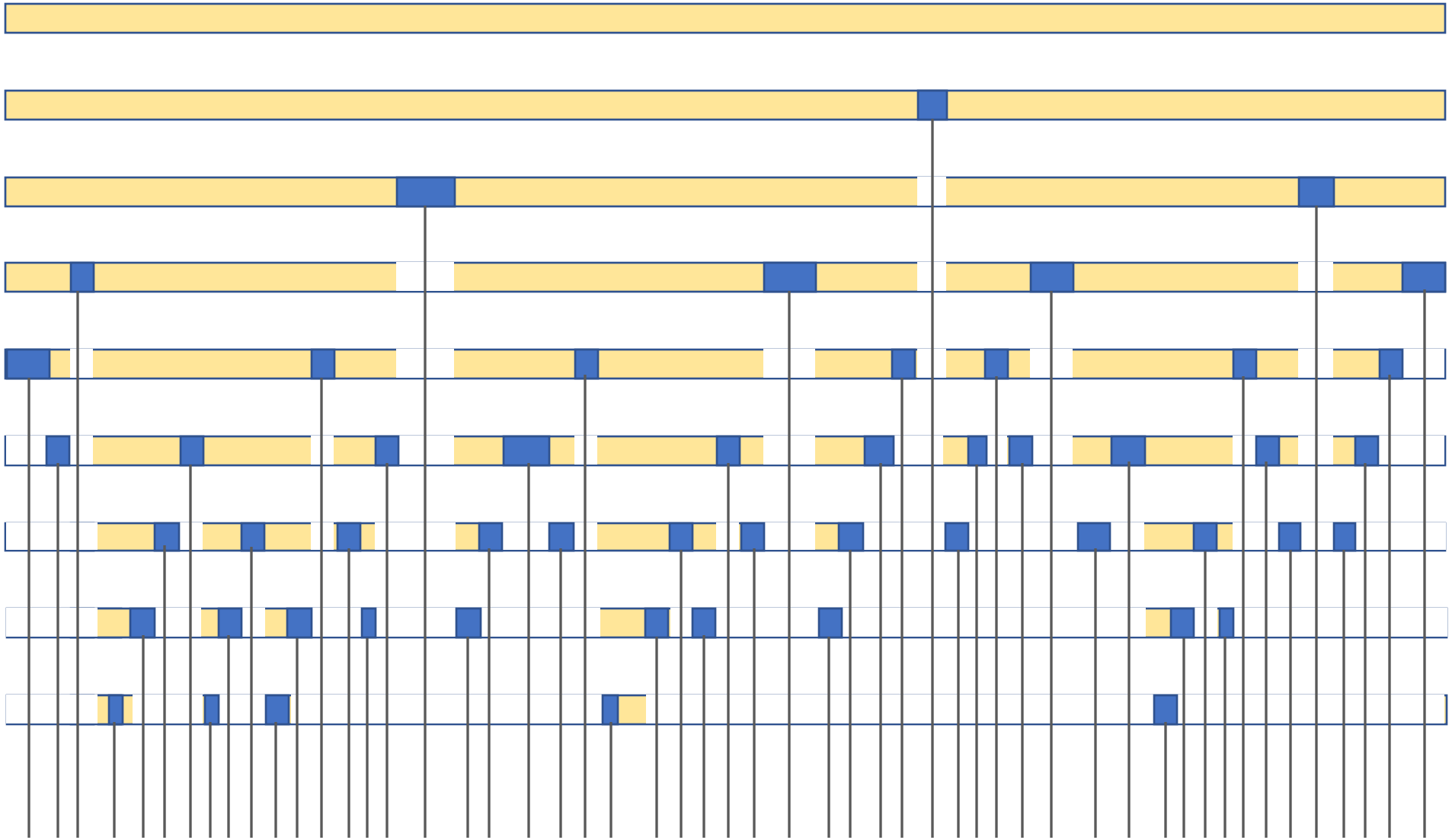


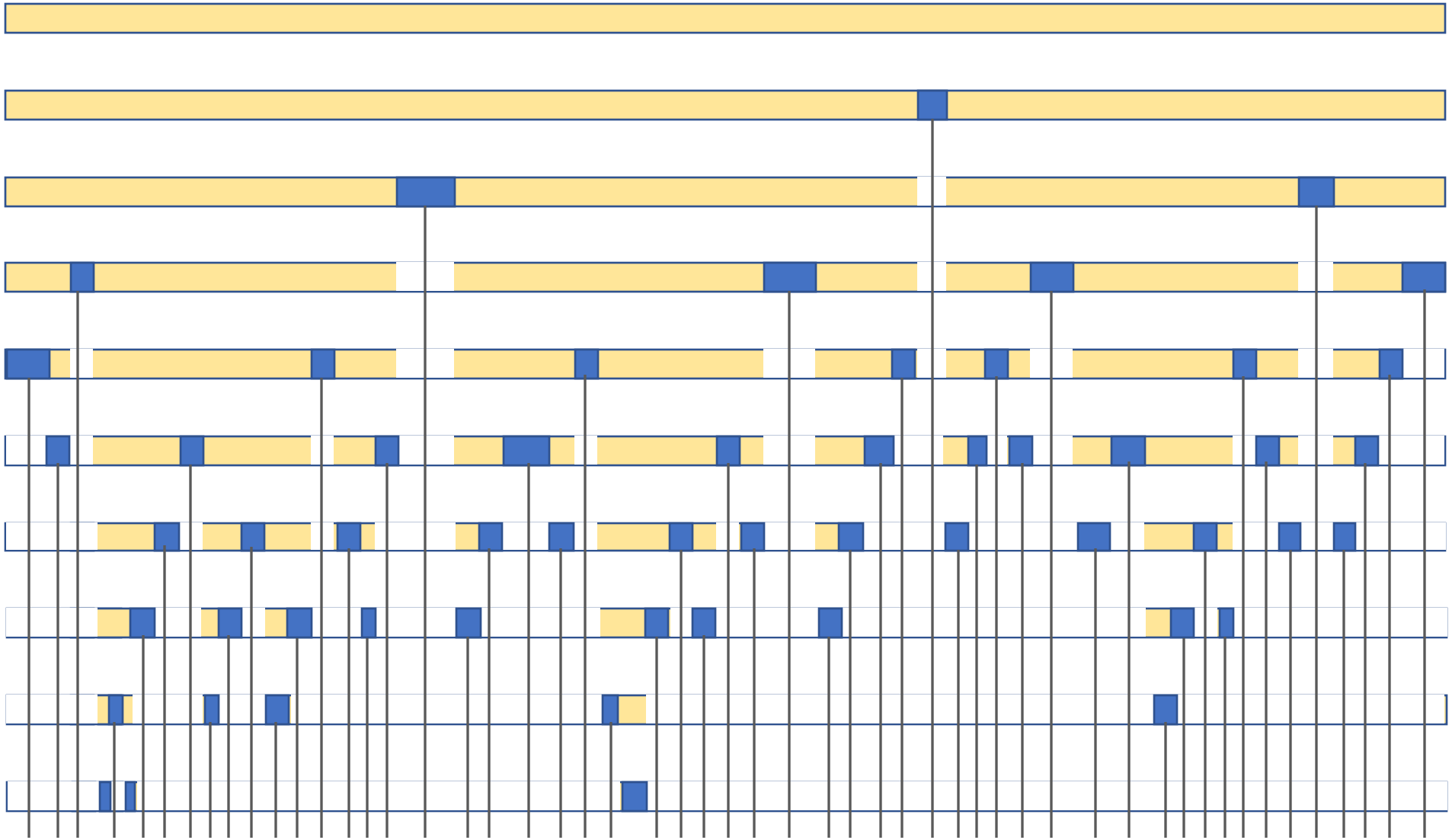


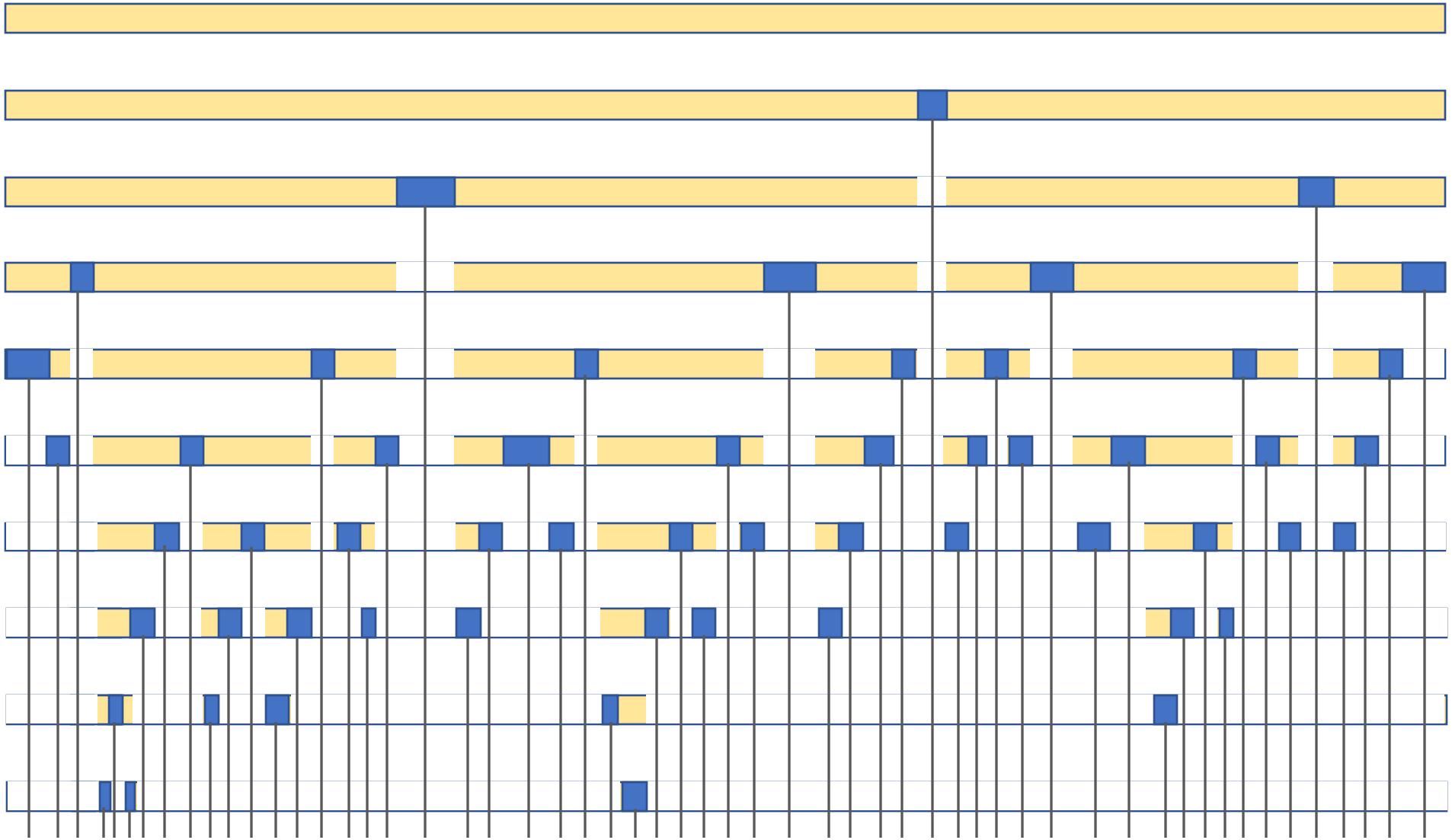


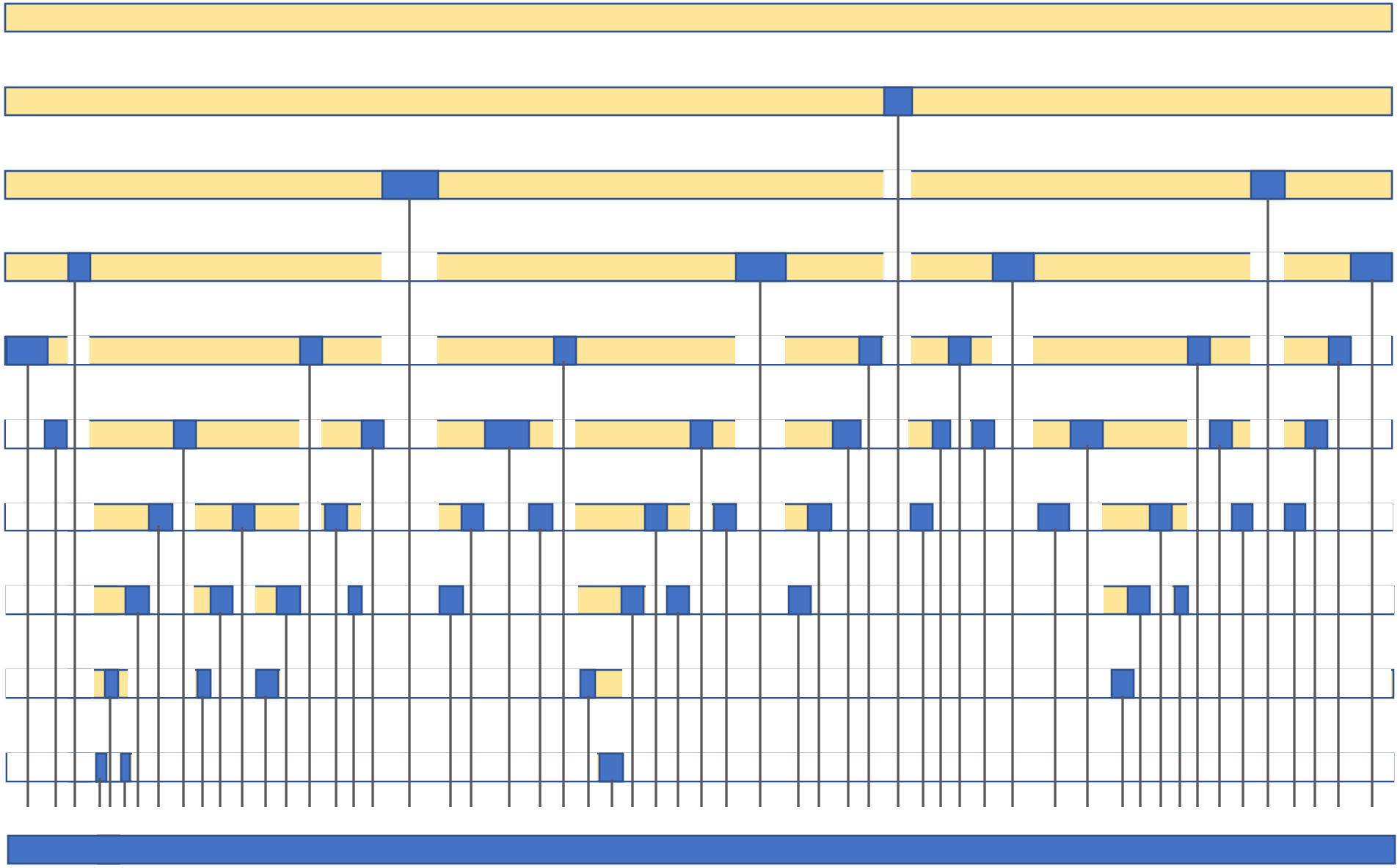


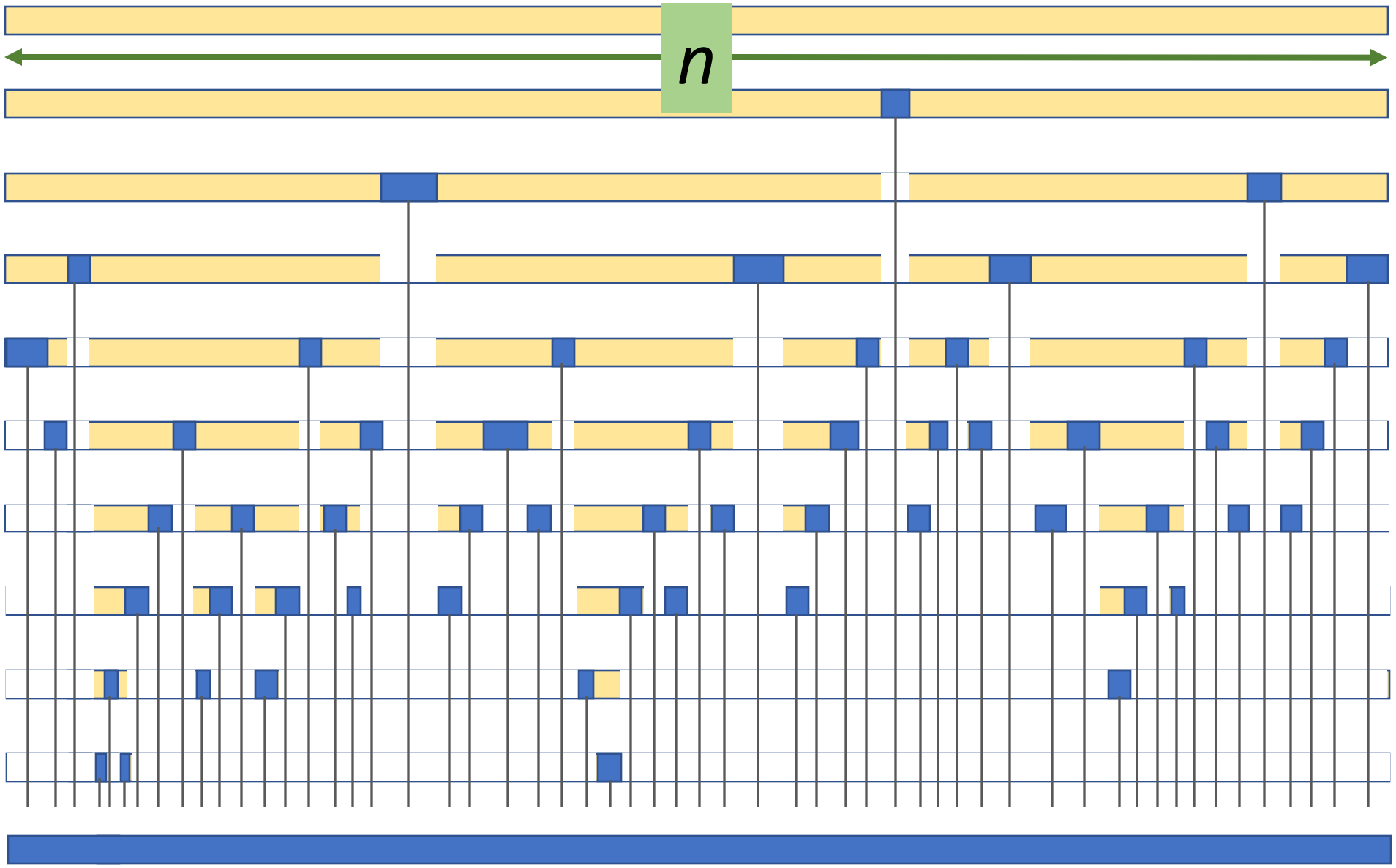








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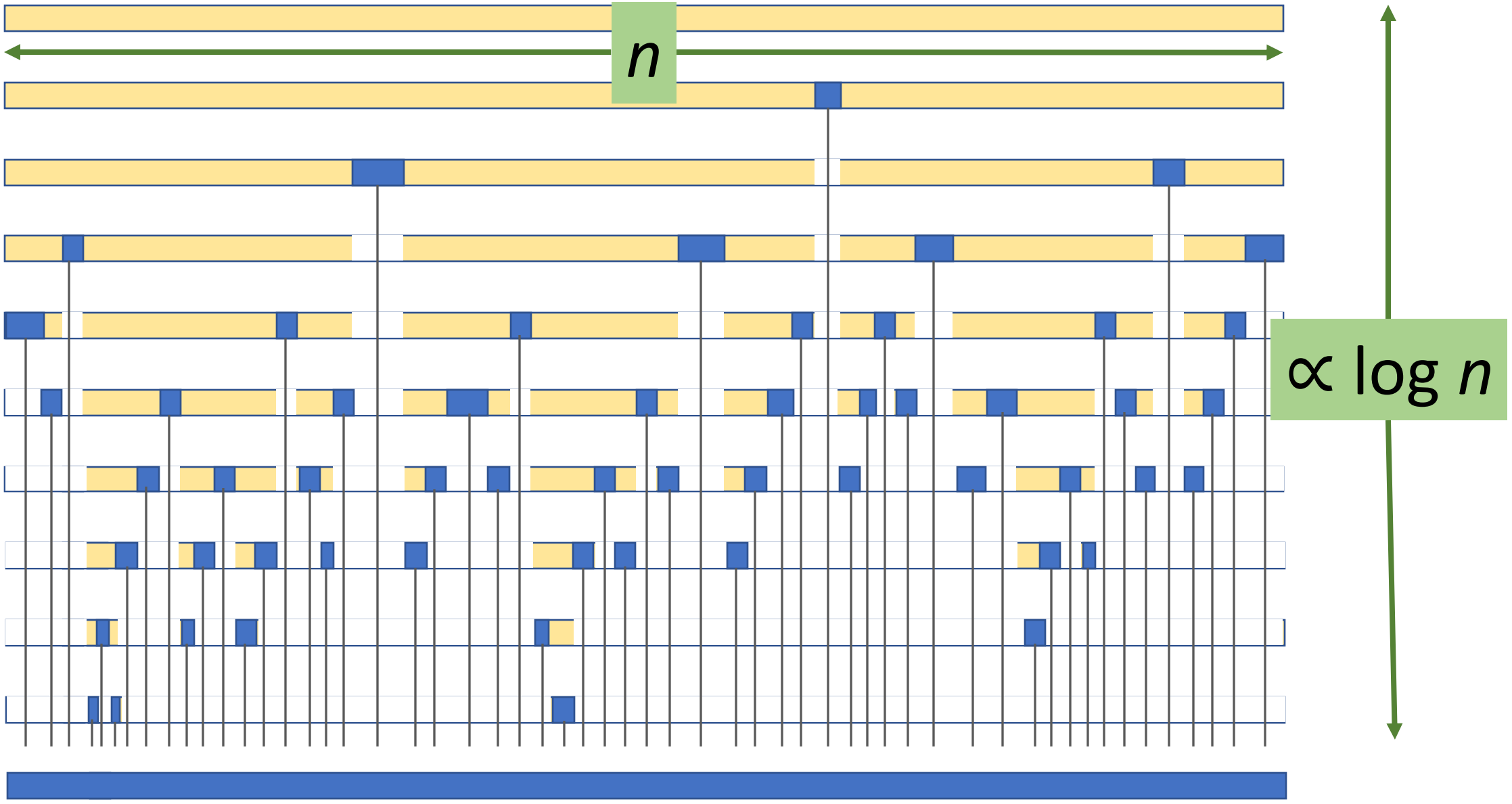
n
∝ log n